ISC Class XII Mathematics

Chapter 3
Matrices

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101 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

Compiled Answer Key – Step-by-step ISC board-style solutions

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Exercise 3.2 – Addition, Subtraction and Scalar Multiplication of Matrices

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2	If $A = \begin{bmatrix}2 & 43 & 2\end{bmatrix}$ ...	7	Numerical	3.2
(i)	$A + B$	7	Numerical	3.2
(ii)	$A - B$	7	Numerical	3.2
(iii)	$3A - C$	7	Numerical	3.2
3	Compute the following:	7	Numerical	3.2
(i)	$\begin{bmatrix}-1 & 4 & -68 & 5 & 162 & 8 & 5\end{bmatrix} + \begin{bmatrix}12 & 7 & 68 & 0 & 53 & 2 & 4\end{bmatrix}$ ...	7	Numerical	3.2
(ii)	$\begin{bmatrix}\cos^2 x & \sin^2 x \sin^2 x & \cos^2 x\end{bmatrix} + \begin{bmatrix}\sin^2 x & \cos^2 x \cos^2 x & \sin^2 x\end{bmatrix}$ ...	7	Numerical	3.2
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5	If $A = \begin{bmatrix}1 & 2 & 32 & 3 & 1\end{bmatrix}$ and ...	7	Numerical	3.2
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10	If ...	5	Numerical	3.2
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12	If ...	5	Numerical	3.2
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Q. No.	Topic / Statement Summary	Type No.	Kind	Section
14	Find matrices X and Y, if $X+Y=\begin{bmatrix}7 & 02 & 5\end{bmatrix}$ and …	8	Numerical	3.2
15	Find matrices X and Y if …	8	Numerical	3.2
16	Find a matrix X such that $3A-2B+X=O$, where …	8	Numerical	3.2
17	If $A=\begin{bmatrix}9 & 17 & 8\end{bmatrix}$ and …	8	Numerical	3.2

Exercise 3.3 – Multiplication of Matrices

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
1	If $A=[-1 \quad 2 \quad -5]$ and $B=\begin{bmatrix}2-17\end{bmatrix}$, write the …	9	Numerical	3.3
2	If $A=\begin{bmatrix}2 & 1 & 44 & 1 & 5\end{bmatrix}$ and …	9	Numerical	3.3
3	If A is a matrix of order 3×3 and B is a matrix of order 2×3, can …	9	Numerical	3.3
4	If $A=\begin{bmatrix}1 & 2-1 & 12 & 3\end{bmatrix}$ …	10	Numerical	3.3
5	If $A=\begin{bmatrix}2 & 43 & 1-1 & 2\end{bmatrix}$ …	10	Numerical	3.3
6	If $\begin{bmatrix}x & 1\end{bmatrix}\begin{bmatrix}1 & 0-2 & 0\end{bmatrix}=O$ …	11	Numerical	3.3
7	If …	11	Numerical	3.3
8	Write the value of $x+y+z$ if …	11	Numerical	3.3
9	Compute the indicated products:	12	Numerical	3.3
(i)	$\begin{bmatrix}1 & -22 & 3\end{bmatrix}\begin{bmatrix}1 & 2 & 32 & 3 & 1\end{bmatrix}$ …	12	Numerical	3.3
(ii)	$\begin{bmatrix}3 & -1 & 3-1 & 0 & 2\end{bmatrix}\begin{bmatrix}2 & -31 & 03 & 1\end{bmatrix}$ …	12	Numerical	3.3
10	If $A=\begin{bmatrix}\alpha & \beta\gamma & -\alpha\end{bmatrix}$ and $A^2=I$, find …	11	Numerical	3.3
11	If $A=\begin{bmatrix}1 & 00 & -1\end{bmatrix}$ and …	12	Numerical	3.3
12	See parts below:	13	Proof	3.3
(i)	If $A=\begin{bmatrix}1 & 11 & 1\end{bmatrix}$ and …	13	Proof	3.3
(ii)	If $A=\begin{bmatrix}1 & 23 & 4\end{bmatrix}$ …	13	Proof	3.3
13	If $A=\begin{bmatrix}\alpha & 01 & 1\end{bmatrix}$ …	11	Numerical	3.3
14	If A is a square matrix such that $A^2=I$, then find the simplified value of …	14	Numerical	3.3
15	Write the following as a single matrix:	15	Numerical	3.3
(i)	$\begin{bmatrix}1 & -2 & 3\end{bmatrix}\begin{bmatrix}2 & -1 & 50 & 2 & 47 & 5 & 0\end{bmatrix}+\begin{bmatrix}2 & -5 & 7\end{bmatrix}$ …	15	Numerical	3.3
(ii)	$\begin{bmatrix}3 & 2 & 57 & -4 & 0\end{bmatrix}\begin{bmatrix}2 & 22 & -13 & 5\end{bmatrix}-\begin{bmatrix}7 & -85 & 9\end{bmatrix}$ …	15	Numerical	3.3
16	See parts below:	16	Proof	3.3
(i)	If …	16	Proof	3.3
(ii)	If …	16	Proof	3.3
17	If $A=\begin{bmatrix}2 & -13 & 2\end{bmatrix}$ and …	15	Numerical	3.3
18	If $A=\begin{bmatrix}2 & 1\end{bmatrix}$ …	13	Proof	3.3
19	See parts below:	11	Numerical	3.3
(i)	Solve for x and y, given that …	11	Numerical	3.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Solve for x and y, given that ...	11	Numerical	3.3
20	See parts below:	17	Proof	3.3
(i)	If $A = \begin{bmatrix}3 & -5 - 4 & 2\end{bmatrix}$, show that ...	17	Proof	3.3
(ii)	If $A = \begin{bmatrix}1 & 24 & 1\end{bmatrix}$, find the value of ...	17	Proof	3.3
21	If $A = \begin{bmatrix}1 & 00 & 1\end{bmatrix}$ and ...	13	Proof	3.3
22	If $A = \begin{bmatrix}4 & 2 - 1 & 1\end{bmatrix}$, prove that $(A - 2I)(A - 3I) = O$.	17	Proof	3.3
23	If $A = \begin{bmatrix}3 & -24 & -2\end{bmatrix}$, then find k so that ...	11	Numerical	3.3
24	*(see parts below)*	17	Proof	3.3
(i)	If $A = \begin{bmatrix}1 & 22 & 1\end{bmatrix}$, show that $f(A) = O$, where ...	17	Proof	3.3
(ii)	If $A = \begin{bmatrix}-1 & 23 & 1\end{bmatrix}$, find $f(A)$, where ...	17	Numerical	3.3
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(i)	find λ, μ so that $A^2 = \lambda A + \mu I$	17	Numerical	3.3
(ii)	prove that $A^3 - 4A^2 + A = O$	17	Numerical	3.3
26	If $A = \begin{bmatrix}1 & 2 & 22 & 1 & 22 & 2 & 1\end{bmatrix}$, verify that ...	17	Proof	3.3
27	See parts below:	13	Proof	3.3
(i)	Let A and B be square matrices of same order. Does ...	13	Proof	3.3
(ii)	Let A and B be square matrices of order 3×3. Is $(AB)^2 = A^2B^2$? ...	13	Proof	3.3
28	If A and B are square matrices of the same order such that $AB = BA$, prove that:	13	Proof	3.3
(i)	$(A + B)(A - B) = A^2 - B^2$	13	Proof	3.3
(ii)	$(A + B)^2 = A^2 + 2AB + B^2$	13	Proof	3.3
(iii)	$(A - B)^2 = A^2 - 2AB + B^2$	13	Proof	3.3
(iv)	$(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$	13	Proof	3.3
(v)	$(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$	13	Proof	3.3
29	If $P = \begin{bmatrix}p & q - q & p\end{bmatrix}$ and ...	13	Proof	3.3
30	Find x, if:	11	Numerical	3.3
(i)	$\begin{bmatrix}1 & x & 1\end{bmatrix} \begin{bmatrix}1 & 3 & 22 & 5 & 115 & 3 & 2\end{bmatrix} \begin{bmatrix}12x\end{bmatrix} = O$ ...	11	Numerical	3.3
(ii)	$\begin{bmatrix}1 & 2 & 1\end{bmatrix} \begin{bmatrix}1 & 2 & 02 & 0 & 11 & 0 & 2\end{bmatrix} \begin{bmatrix}02x\end{bmatrix} = O$ ...	11	Numerical	3.3
31	If $A = \begin{bmatrix}2 & 3 - 1 & 2\end{bmatrix}$ and $f(x) = x^2 - 4x + 7$, show that ...	18	Numerical	3.3
32	If $A = \begin{bmatrix}1 & 11 & 1\end{bmatrix}$, prove that ...	19	Proof	3.3
33	See parts below:	20	Numerical	3.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Find a 2×2 matrix B such that ...	20	Numerical	3.3
(ii)	If $A = \begin{bmatrix} 3 & -4 - 1 & 2 \end{bmatrix}$, find matrix B such that ...	20	Numerical	3.3
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35	Three students Ram, Mohan and Ankit go to a shop to buy stationary. Ram purchases 2 dozen ...	21	Numerical	3.3
Exercise 3.4 – Transpose, Symmetric and Skew-Symmetric Matrices				
1	If $A = \begin{bmatrix} 2 & -3 & 0 - 1 & 4 & 5 \end{bmatrix}$, then find $(3A)'$.	22	Numerical	3.4
2	If $A = \begin{bmatrix} 2 & -1 & 54 & 0 & 3 \end{bmatrix}$ and ...	22	Numerical	3.4
3	If ...	22	Numerical	3.4
4	If $A^T = \begin{bmatrix} 3 & 4 - 1 & 20 & 1 \end{bmatrix}$ and ...	22	Numerical	3.4
5	For what value of k the matrix $\begin{bmatrix} 0 & k - 6 & 0 \end{bmatrix}$ is a ...	23	Numerical	3.4
6	*(see parts below)*	23	Proof	3.4
(i)	Show that $(A + A')$ is a symmetric matrix, if ...	23	Proof	3.4
(ii)	If $A = \begin{bmatrix} 5 & -1 - 2 & 6 \end{bmatrix}$, determine whether $A - A^T$ is ...	23	Numerical	3.4
7	*(see parts below)*	23	Proof	3.4
(i)	If $A = \begin{bmatrix} 5 & ab & 0 \end{bmatrix}$ and A is symmetric matrix, show that ...	23	Proof	3.4
(ii)	If $A = \begin{bmatrix} 5 & xy & 0 \end{bmatrix}$ and A is symmetric, then write the ...	23	Numerical	3.4
(iii)	If the matrix $\begin{bmatrix} 6 & -x^2 2x - 15 & 10 \end{bmatrix}$ is symmetric, find the ...	23	Numerical	3.4
8	If A is a square matrix, prove that $A'A$ is symmetric.	24	Proof	3.4
9	If $A = \begin{bmatrix} 3 & 57 & 9 \end{bmatrix}$ is written as $A = P + Q$, where P is ...	25	Numerical	3.4
10	If A, B are symmetric matrices and $AB = BA$, then show that AB is symmetric.	24	Proof	3.4
11	If A, B and AB are all symmetric matrices, then show that $AB = BA$.	24	Proof	3.4
12	If A, B are skew-symmetric matrices and $AB = BA$, then show that AB is symmetric.	24	Proof	3.4
13	If A, B are skew-symmetric matrices of same order and AB is symmetric, then show that ...	24	Proof	3.4
14	Let $A = \begin{bmatrix} 5 & -16 & 7 \end{bmatrix}$ and ...	22	Proof	3.4
15	If $A = \begin{bmatrix} 3 & \sqrt{3} & 24 & 2 & 0 \end{bmatrix}$ and ...	22	Proof	3.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
16	If $A' = \begin{bmatrix}-2 & 31 & 2\end{bmatrix}$ and ...	22	Numerical	3.4
17	If $A = \begin{bmatrix}1-43\end{bmatrix}$ and ...	22	Proof	3.4
18	If $A = \begin{bmatrix}1 & 24 & 15 & 6\end{bmatrix}$ and ...	22	Proof	3.4
19	If $A = \begin{bmatrix}2 & 4 & 03 & 9 & 6\end{bmatrix}$ and ...	22	Proof	3.4
20	If $A = \begin{bmatrix}3 & 2-1 & 1\end{bmatrix}$ and ...	22	Numerical	3.4
21	If $A = \begin{bmatrix}\sin\alpha & \cos\alpha - \cos\alpha & \sin\alpha\end{bmatrix}$ ...	26	Proof	3.4
22	If $A = \frac{1}{3}\begin{bmatrix}1 & -2 & 2-2 & 1 & 2-2 & -2 & -1\end{bmatrix}$ ...	26	Proof	3.4
23	If $A = \begin{bmatrix}1 & 2 & 22 & 1 & x-2 & 2 & -1\end{bmatrix}$ is a matrix ...	26	Numerical	3.4
24	Find the integral value of x if ...	22	Numerical	3.4
25	Show that the matrix A, where ...	23	Proof	3.4
26	If the matrix $\begin{bmatrix}0 & 2b & -23 & 1 & 33a & 3 & -1\end{bmatrix}$ is ...	23	Numerical	3.4
27	If A and B are symmetric matrices of the same order, prove that $AB + BA$ is ...	24	Proof	3.4
28	If A and B are symmetric matrices such that AB and BA are both defined, then prove ...	24	Proof	3.4
29	Express each of the following matrices as the sum of a symmetric and a skew-symmetric ...	25	Numerical	3.4
(i)	$\begin{bmatrix}3 & 51 & -1\end{bmatrix}$	25	Numerical	3.4
(ii)	$\begin{bmatrix}3 & -41 & -1\end{bmatrix}$	25	Numerical	3.4
30	Express the following matrices as the sum of symmetric and skew-symmetric matrices:	25	Numerical	3.4
(i)	$\begin{bmatrix}3 & 2 & 54 & 1 & 30 & 6 & 7\end{bmatrix}$	25	Numerical	3.4
(ii)	$\begin{bmatrix}2 & 4 & -67 & 3 & 51 & -2 & 4\end{bmatrix}$	25	Numerical	3.4
31	For what value of x, is ...	23	Numerical	3.4
32	If the matrix $A = \begin{bmatrix}a & 6 & b2c & x+1 & 8-1 & -8 & y-3\end{bmatrix}$ is ...	23	Numerical	3.4

Formula & Standard Results Sheet

I. Order, Elements and Construction of Matrices (Ex 3.1)

Order of a Matrix

$$A_{m\times n} \implies m \text{ rows}, \; n \text{ columns}$$

The order is always stated as rows × columns

General Element Notation

$$A = [a_{ij}]_{m\times n}$$

a_{ij} denotes the entry in the i-th row and j-th column

Number of Possible Matrices

$$\text{Total matrices of order } m\times n \text{ from } k \text{ available entries} = k^{m\times n}$$

Each of the mn positions can independently take any of the k values

Equality of Matrices

$$A = [a_{ij}] = B = [b_{ij}] \iff \text{same order and } a_{ij} = b_{ij} \; \forall\, i, j$$

Both the order and every corresponding entry must match

II. Addition, Subtraction and Scalar Multiplication (Ex 3.2)

Matrix Addition and Subtraction

$$A \pm B = [a_{ij} \pm b_{ij}]$$

Defined only when A and B have the same order

Scalar Multiplication

$$kA = [k\, a_{ij}]$$

Every entry of A is multiplied by the scalar k

Linear Combination of Matrices

$$kA + mB = [k\, a_{ij} + m\, b_{ij}]$$

Combines scalar multiplication and addition entry-by-entry

Solving Simultaneous Matrix Equations

$$X + Y = A, \; X - Y = B \implies 2X = A + B, \;\; 2Y = A - B$$

Add and subtract the two equations, then divide by the scalar 2, exactly as with ordinary simultaneous equations

III. Multiplication of Matrices (Ex 3.3)

Compatibility for Multiplication

$$P_{m\times n}\cdot Q_{n\times p}=(PQ)_{m\times p}$$

The number of columns of P must equal the number of rows of Q; the product takes the outer dimensions

Element of a Product Matrix

$$(AB)_{ij}=\sum_k a_{ik}\,b_{kj}$$

Row i of A dotted with column j of B

Identity Matrix Property

$$AI=IA=A$$

Multiplying by the identity matrix leaves A unchanged

Distributive Property

$$A(B+C)=AB+AC$$

Matrix multiplication distributes over addition

Associative Property

$$(AB)C=A(BC)$$

Grouping does not affect the product

Non-Commutativity of Multiplication

$$AB\neq BA\quad\text{(in general)}$$

Always verify the order of multiplication; do not assume $AB=BA$ unless proved

Matrix Polynomial

$$f(x)=a_nx^n+\cdots+a_1x+a_0\implies f(A)=a_nA^n+\cdots+a_1A+a_0I$$

The constant term becomes a multiple of the identity matrix, not a bare number

Reducing Higher Powers via a Matrix Relation

If $A^2=kA-mI$, use it repeatedly to express A^n in terms of A and I

Avoids computing high powers of A directly by hand

Principle of Mathematical Induction (on Matrices)

$$P(1) \text{ true, and } P(k) \implies P(k+1) \implies P(n) \text{ true } \forall n \in \mathbb{N}$$

Standard three-step structure: base case, inductive hypothesis, inductive step

IV. Transpose, Symmetric and Skew-Symmetric Matrices (Ex 3.4)

Transpose of a Matrix

$$(A')_{ij} = A_{ji}$$

If A is $m \times n$, then A' is $n \times m$; rows become columns

Properties of Transpose

$$(A')' = A, \quad (kA)' = kA', \quad (A \pm B)' = A' \pm B', \quad (AB)' = B'A'$$

Note the reversal of order in $(AB)' = B'A'$

Symmetric Matrix

$$A' = A \iff a_{ij} = a_{ji} \, \forall \, i, j$$

A symmetric matrix must be square

Skew-Symmetric Matrix

$$A' = -A \iff a_{ij} = -a_{ji}, \quad \text{so } a_{ii} = 0$$

Every diagonal entry of a skew-symmetric matrix is 0

Symmetric-Skew-Symmetric Decomposition

$$A = P + Q, \quad P = \frac{1}{2}(A + A') \text{ (symmetric)}, \quad Q = \frac{1}{2}(A - A') \text{ (skew-symmetric)}$$

Every square matrix can be written uniquely as this sum

Orthogonal Matrix

$$A \text{ is orthogonal} \iff AA' = I$$

Equivalently, $A' = A^{-1}$

3.1 – Order, Elements and Construction of Matrices

Question 1

If A is a matrix of type $p \times q$ and R is a row of A, then what is the type of R as a matrix?

TYPE 1 Matrix Order Analysis

Given: A is a matrix of order $p \times q$ and R is a row matrix of A.

Concept / Formula Used: n $1 \times n$

Solution:

Since A is a matrix of order $p \times q$, it has p rows and q columns.

Each row R of matrix A contains q elements corresponding to the q columns of A.

A single row of A, treated as a standalone matrix R, has 1 row and q columns.

Therefore, the type (order) of R as a matrix is $1 \times q$.

Final Answer

The type of R as a matrix is $1 \times q$.

Question 2

If A is a column matrix with 5 rows, then what type of matrix is a row of A?

TYPE 1 Matrix Order Analysis

Given: A is a column matrix with 5 rows.

Concept / Formula Used: m $m \times 1$ 1×1

Solution:

Since A is a column matrix with 5 rows, the order of matrix A is 5×1.

Matrix A has only 1 column.

A row of matrix A contains elements across all columns of A. Since there is only 1 column in A, a row of A contains only 1 element.

A row of A, viewed as a matrix itself, has 1 row and 1 column.

Therefore, the order (type) of a row of A as a matrix is 1×1.

Final Answer

A row of A is a 1×1 matrix.

Question 3

If a matrix has 18 elements, what the possible orders it can have? What, if it has 11 elements?

TYPE 2 **Possible Matrix Orders from Total Elements**

Given: Number of elements in a matrix is 18, or 11.

Concept / Formula Used: $n \quad (m, k) \quad m \times k = n$

Solution:

Case I: Matrix has 18 elements

We need to find all ordered pairs (m, k) of positive integers such that $m \times k = 18$.

The factor pairs of 18 are:

$$\begin{aligned} 1 \times 18 &= 18 \\ 2 \times 9 &= 18 \\ 3 \times 6 &= 18 \\ 6 \times 3 &= 18 \\ 9 \times 2 &= 18 \\ 18 \times 1 &= 18 \end{aligned}$$

Hence, the possible orders are $1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 18 \times 1$.

Case II: Matrix has 11 elements

We need to find all ordered pairs (m, k) of positive integers such that $m \times k = 11$.

Since 11 is a prime number, its only positive integral factor pairs are:

$$\begin{aligned} 1 \times 11 &= 11 \\ 11 \times 1 &= 11 \end{aligned}$$

Hence, the possible orders are $1 \times 11, 11 \times 1$.

Final Answer

For 18 elements: $1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 18 \times 1$.

For 11 elements: $1 \times 11, 11 \times 1$.

Question 4

(i) If A is a 3×3 matrix whose elements are given by $a_{ij} = \frac{1}{3}|-3i + j|$, write the value of a_{23}.

(ii) Write the element a_{12} of the matrix $A = [a_{ij}]_{2\times 2}$, whose elements a_{ij} are given by $a_{ij} = 2^{2ix} \sin jx$.

TYPE 3 **Evaluating Specific Matrix Elements**

Given: (i) $a_{ij} = \frac{1}{3}|-3i + j|$ for $A_{3\times3}$.
(ii) $a_{ij} = 2^{2ix}\sin(jx)$ for $A_{2\times2}$.

Concept / Formula Used: a_{ij} i j a_{ij}

Solution:

Part (i):

We need to find a_{23}, so set $i = 2$ and $j = 3$.

Substitute $i = 2$ and $j = 3$ into $a_{ij} = \frac{1}{3}|-3i + j|$:

$$\begin{aligned} a_{23} &= \frac{1}{3}|-3(2) + 3| \\ &= \frac{1}{3}|-6 + 3| \\ &= \frac{1}{3}|-3| \\ &= \frac{1}{3} \times 3 \\ &= 1 \end{aligned}$$

Part (ii):

We need to find a_{12}, so set $i = 1$ and $j = 2$.

Substitute $i = 1$ and $j = 2$ into $a_{ij} = 2^{2ix}\sin(jx)$:

$$\begin{aligned} a_{12} &= 2^{2(1)x}\sin(2x) \\ &= 2^{2x}\sin(2x) \end{aligned}$$

Final Answer

(i) $a_{23} = 1$

(ii) $a_{12} = 2^{2x}\sin(2x)$

Question 5

(i) For a 2×2 matrix, $A = [a_{ij}]$, whose elements are given by $a_{ij} = \frac{i}{j}$, write the value of a_{12}.

(ii) Write the element a_{23} of a 3×3 matrix $A = [a_{ij}]$ whose elements a_{ij} are given by $a_{ij} = \frac{|i-j|}{2}$.

TYPE 3 **Evaluating Specific Matrix Elements**

Given: (i) $a_{ij} = \frac{i}{j}$ for $A_{2\times 2}$.

(ii) $a_{ij} = \frac{|i-j|}{2}$ for $A_{3\times 3}$.

Concept / Formula Used: i j i j a_{ij}

Solution:

Part (i):

To find a_{12}, substitute $i = 1$ and $j = 2$ into $a_{ij} = \frac{i}{j}$:

$$a_{12} = \frac{1}{2}$$

Part (ii):

To find a_{23}, substitute $i = 2$ and $j = 3$ into $a_{ij} = \frac{|i-j|}{2}$:

$$\begin{aligned} a_{23} &= \frac{|2-3|}{2} \\ &= \frac{|-1|}{2} \\ &= \frac{1}{2} \end{aligned}$$

Final Answer

(i) $a_{12} = \frac{1}{2}$

(ii) $a_{23} = \frac{1}{2}$

Question 6

Construct a 2×2 matrix $A = [a_{ij}]$ whose elements a_{ij} are given by:
(i) $a_{ij} = 2i - j$
(ii) $a_{ij} = |2i - 3j|$

TYPE 4 **Matrix Construction from General Element**

Given: $A = [a_{ij}]$ of order 2×2, where (i) $a_{ij} = 2i - j$, (ii) $a_{ij} = |2i - 3j|$.

Concept / Formula Used: 2×2 $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

Solution:

Part (i): $a_{ij} = 2i - j$

Evaluating elements for $i = 1, 2$ and $j = 1, 2$:

$$\begin{aligned} a_{11} &= 2(1) - 1 = 2 - 1 = 1 \\ a_{12} &= 2(1) - 2 = 2 - 2 = 0 \\ a_{21} &= 2(2) - 1 = 4 - 1 = 3 \\ a_{22} &= 2(2) - 2 = 4 - 2 = 2 \end{aligned}$$

Thus, $A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$.

Part (ii): $a_{ij} = |2i - 3j|$

Evaluating elements for $i = 1, 2$ and $j = 1, 2$:

$$\begin{aligned} a_{11} &= |2(1) - 3(1)| = |2 - 3| = |-1| = 1 \\ a_{12} &= |2(1) - 3(2)| = |2 - 6| = |-4| = 4 \\ a_{21} &= |2(2) - 3(1)| = |4 - 3| = |1| = 1 \\ a_{22} &= |2(2) - 3(2)| = |4 - 6| = |-2| = 2 \end{aligned}$$

Thus, $A = \begin{bmatrix} 1 & 4 \\ 1 & 2 \end{bmatrix}$.

Final Answer

(i) $A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$

(ii) $A = \begin{bmatrix} 1 & 4 \\ 1 & 2 \end{bmatrix}$

Question 7

Construct a 2×3 matrix $B = [b_{ij}]$ whose elements b_{ij} are given by:

(i) $b_{ij} = i - 3j$

(ii) $b_{ij} = (i + 2j)^2$

TYPE 4 Matrix Construction from General Element

Given: $B = [b_{ij}]$ of order 2×3, where (i) $b_{ij} = i - 3j$, (ii) $b_{ij} = (i + 2j)^2$.

Concept / Formula Used: 2×3 $\quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$

Solution:

Part (i): $b_{ij} = i - 3j$

Evaluating elements for $i = 1, 2$ and $j = 1, 2, 3$:

$$\begin{aligned}
b_{11} &= 1 - 3(1) = 1 - 3 = -2 \\
b_{12} &= 1 - 3(2) = 1 - 6 = -5 \\
b_{13} &= 1 - 3(3) = 1 - 9 = -8 \\
b_{21} &= 2 - 3(1) = 2 - 3 = -1 \\
b_{22} &= 2 - 3(2) = 2 - 6 = -4 \\
b_{23} &= 2 - 3(3) = 2 - 9 = -7
\end{aligned}$$

Thus, $B = \begin{bmatrix} -2 & -5 & -8 \\ -1 & -4 & -7 \end{bmatrix}$.

Part (ii): $b_{ij} = (i + 2j)^2$

Evaluating elements for $i = 1, 2$ and $j = 1, 2, 3$:

$$\begin{aligned}
b_{11} &= (1 + 2(1))^2 = (1 + 2)^2 = 3^2 = 9 \\
b_{12} &= (1 + 2(2))^2 = (1 + 4)^2 = 5^2 = 25 \\
b_{13} &= (1 + 2(3))^2 = (1 + 6)^2 = 7^2 = 49 \\
b_{21} &= (2 + 2(1))^2 = (2 + 2)^2 = 4^2 = 16 \\
b_{22} &= (2 + 2(2))^2 = (2 + 4)^2 = 6^2 = 36 \\
b_{23} &= (2 + 2(3))^2 = (2 + 6)^2 = 8^2 = 64
\end{aligned}$$

Thus, $B = \begin{bmatrix} 9 & 25 & 49 \\ 16 & 36 & 64 \end{bmatrix}$.

Final Answer

(i) $B = \begin{bmatrix} -2 & -5 & -8 \\ -1 & -4 & -7 \end{bmatrix}$

(ii) $B = \begin{bmatrix} 9 & 25 & 49 \\ 16 & 36 & 64 \end{bmatrix}$

Question 8

(i) If $\begin{bmatrix} x + 2y & -y \\ 3x & 4 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 6 & 4 \end{bmatrix}$, find the values of x and y.

(ii) If $\begin{bmatrix} x + 3y & y \\ 7 - x & 4 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 0 & 4 \end{bmatrix}$, find the values of x and y.

(iii) Find the value of y, if $\begin{bmatrix} x - y & 2 \\ x & 5 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix}$.

(iv) Find the value of x, if $\begin{bmatrix} 3x + y & -y \\ 2y - x & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -5 & 3 \end{bmatrix}$.

(v) If $\begin{bmatrix} x + 2y & 3y \\ 4x & 2 \end{bmatrix} = \begin{bmatrix} 0 & -3 \\ 8 & 2 \end{bmatrix}$, find the value of $(x - y)$.

(vi) If $\begin{bmatrix} x - y & z \\ 2x - y & w \end{bmatrix} = \begin{bmatrix} -1 & 4 \\ 0 & 5 \end{bmatrix}$, find the value of $(x + y)$.

(vii) If $\begin{bmatrix} x+y & x+2 \\ 2x-y & 16 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ 1 & 3y+1 \end{bmatrix}$, then write the value of $(y-x)$.

TYPE 5 Equality of Matrices

Given: Seven pairs of equal 2×2 matrices with unknown variable entries.

Concept / Formula Used: $A = [a_{ij}] \qquad B = [b_{ij}] \qquad a_{ij} = b_{ij} \qquad i, j$

Solution:

Part (i):

Equating corresponding elements:

$$\begin{aligned} -y = 3 &\implies y = -3 \\ 3x = 6 &\implies x = 2 \end{aligned}$$

Part (ii):

Equating corresponding elements:

$$\begin{aligned} y &= -1 \\ 7 - x = 0 &\implies x = 7 \end{aligned}$$

Part (iii):

Equating corresponding elements:

$$\begin{aligned} x &= 3 \\ x - y &= 2 \end{aligned}$$

Substitute $x = 3$:

$$\begin{aligned} 3 - y &= 2 \\ y &= 1 \end{aligned}$$

Part (iv):

Equating corresponding elements:

$$\begin{aligned} -y = 2 &\implies y = -2 \\ 3x + y = 1 & \end{aligned}$$

Substitute $y = -2$:

$$\begin{aligned} 3x + (-2) &= 1 \\ 3x &= 3 \\ x &= 1 \end{aligned}$$

Part (v):

Equating corresponding elements:

$$3y = -3 \implies y = -1$$
$$4x = 8 \implies x = 2$$

Now compute $x - y$:

$$\begin{aligned} x - y &= 2 - (-1) \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

Part (vi):

Equating corresponding elements:

$$x - y = -1 \quad \text{— (1)}$$
$$2x - y = 0 \quad \text{— (2)}$$

Subtracting equation (1) from equation (2):

$$\begin{aligned} (2x - y) - (x - y) &= 0 - (-1) \\ x &= 1 \end{aligned}$$

Substitute $x = 1$ into equation (1):

$$\begin{aligned} 1 - y &= -1 \\ y &= 2 \end{aligned}$$

Now compute $x + y$:

$$\begin{aligned} x + y &= 1 + 2 \\ &= 3 \end{aligned}$$

Part (vii):

Equating corresponding elements:

$$x + 2 = 5 \implies x = 3$$
$$3y + 1 = 16 \implies 3y = 15 \implies y = 5$$

Now compute $y - x$:

$$\begin{aligned} y - x &= 5 - 3 \\ &= 2 \end{aligned}$$

Final Answer

(i) $x = 2, y = -3$
(ii) $x = 7, y = -1$
(iii) $y = 1$
(iv) $x = 1$
(v) $x - y = 3$
(vi) $x + y = 3$
(vii) $y - x = 2$

Question 9

Write the value of $x - y + z$ from the following equation:

$$\begin{bmatrix} x+y+z \\ x+z \\ y+z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$$

TYPE 5 Equality of Matrices

Given: $\begin{bmatrix} x+y+z \\ x+z \\ y+z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$

Solution:

Equating corresponding elements:

$$\begin{aligned} x+y+z &= 9 \quad \text{— (1)} \\ x+z &= 5 \quad \text{— (2)} \\ y+z &= 7 \quad \text{— (3)} \end{aligned}$$

Substitute equation (3) into equation (1):

$$\begin{aligned} x+(y+z) &= 9 \\ x+7 &= 9 \\ x &= 9-7 \\ x &= 2 \end{aligned}$$

Substitute $x = 2$ into equation (2):

$$\begin{aligned} 2+z &= 5 \\ z &= 5-2 \\ z &= 3 \end{aligned}$$

Substitute $z = 3$ into equation (3):

$$\begin{aligned} y+3 &= 7 \\ y &= 7-3 \\ y &= 4 \end{aligned}$$

Now evaluate $x - y + z$:

$$\begin{aligned} x-y+z &= 2-4+3 \\ &= 5-4 \\ &= 1 \end{aligned}$$

Final Answer

$x - y + z = 1$

Question 10

If $\begin{bmatrix} x & 3x-y \\ 2x+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, find x, y, z and w.

TYPE 5 Equality of Matrices

Given: $\begin{bmatrix} x & 3x-y \\ 2x+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$

Solution:

Equating corresponding elements:

$$\begin{aligned} x &= 3 \quad \text{— (1)} \\ 3x - y &= 2 \quad \text{— (2)} \\ 2x + z &= 4 \quad \text{— (3)} \\ 3y - w &= 7 \quad \text{— (4)} \end{aligned}$$

From equation (1):

$$x = 3$$

Substitute $x = 3$ into equation (2):

$$\begin{aligned} 3(3) - y &= 2 \\ 9 - y &= 2 \\ y &= 9 - 2 \\ y &= 7 \end{aligned}$$

Substitute $x = 3$ into equation (3):

$$\begin{aligned} 2(3) + z &= 4 \\ 6 + z &= 4 \\ z &= 4 - 6 \\ z &= -2 \end{aligned}$$

Substitute $y = 7$ into equation (4):

$$\begin{aligned} 3(7) - w &= 7 \\ 21 - w &= 7 \\ w &= 21 - 7 \\ w &= 14 \end{aligned}$$

Final Answer

$x = 3, \quad y = 7, \quad z = -2, \quad w = 14$

Question 11

Find the values of a, b, c and d from the following equations:

$$\begin{bmatrix} 2a+b & a-2b \\ 5c-d & 4c+3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$$

TYPE 5 **Equality of Matrices**

★ **Likely Exam Question**

Given: $\begin{bmatrix} 2a+b & a-2b \\ 5c-d & 4c+3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$

Solution:

Equating corresponding elements, we obtain two independent systems of equations:

System 1 (in a and b):

$$\begin{aligned} 2a+b &= 4 \quad \text{— (1)} \\ a-2b &= -3 \quad \text{— (2)} \end{aligned}$$

Multiply equation (1) by 2:

$$4a+2b = 8 \quad \text{— (3)}$$

Add equation (2) and equation (3):

$$\begin{aligned} (a-2b)+(4a+2b) &= -3+8 \\ 5a &= 5 \\ a &= 1 \end{aligned}$$

Substitute $a = 1$ into equation (1):

$$\begin{aligned} 2(1)+b &= 4 \\ 2+b &= 4 \\ b &= 2 \end{aligned}$$

System 2 (in c and d):

$$\begin{aligned} 5c-d &= 11 \quad \text{— (4)} \\ 4c+3d &= 24 \quad \text{— (5)} \end{aligned}$$

From equation (4), express d in terms of c:

$$d = 5c-11 \quad \text{— (6)}$$

Substitute equation (6) into equation (5):

$$\begin{aligned} 4c + 3(5c - 11) &= 24 \\ 4c + 15c - 33 &= 24 \\ 19c - 33 &= 24 \\ 19c &= 24 + 33 \\ 19c &= 57 \\ c &= \frac{57}{19} \\ c &= 3 \end{aligned}$$

Substitute $c = 3$ into equation (6):

$$\begin{aligned} d &= 5(3) - 11 \\ d &= 15 - 11 \\ d &= 4 \end{aligned}$$

Final Answer

$a = 1, \quad b = 2, \quad c = 3, \quad d = 4$

Question 12

Find x, y, a and b if $\begin{bmatrix} 2x - 3y & a - b & 3 \\ 1 & x + 4y & 3a + 4b \end{bmatrix} = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 6 & 29 \end{bmatrix}$

TYPE 5 **Equality of Matrices**

★ Likely Exam Question

Given: $\begin{bmatrix} 2x - 3y & a - b & 3 \\ 1 & x + 4y & 3a + 4b \end{bmatrix} = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 6 & 29 \end{bmatrix}$

Concept / Formula Used: 2×3

Solution:

By equality of matrices, equate the corresponding entries:

System 1 (in x and y):

$$\begin{aligned} 2x - 3y &= 1 \quad \text{— (1)} \\ x + 4y &= 6 \quad \text{— (2)} \end{aligned}$$

From equation (2), express x in terms of y:

$$x = 6 - 4y \quad \text{— (3)}$$

Substitute equation (3) into equation (1):

$$\begin{aligned} 2(6-4y)-3y &= 1 \\ 12-8y-3y &= 1 \\ 12-11y &= 1 \\ 11y &= 12-1 \\ 11y &= 11 \\ y &= 1 \end{aligned}$$

Substitute $y = 1$ into equation (3):

$$\begin{aligned} x &= 6-4(1) \\ x &= 6-4 \\ x &= 2 \end{aligned}$$

System 2 (in a and b):

$$\begin{aligned} a-b &= -2 \quad \text{— (4)} \\ 3a+4b &= 29 \quad \text{— (5)} \end{aligned}$$

From equation (4), express a in terms of b:

$$a = b-2 \quad \text{— (6)}$$

Substitute equation (6) into equation (5):

$$\begin{aligned} 3(b-2)+4b &= 29 \\ 3b-6+4b &= 29 \\ 7b-6 &= 29 \\ 7b &= 29+6 \\ 7b &= 35 \\ b &= \frac{35}{7} \\ b &= 5 \end{aligned}$$

Substitute $b = 5$ into equation (6):

$$\begin{aligned} a &= 5-2 \\ a &= 3 \end{aligned}$$

Final Answer

$x = 2, \quad y = 1, \quad a = 3, \quad b = 5$

Question 13

(i) Write the number of all possible matrices of order 2×3 with each entry 1 or 2.

(ii) Write the number of all possible matrices of order 2×2 with each entry 1, 2 or 3.

TYPE 6 **Number of Possible Matrices**

To Find: The total number of possible matrices under the given conditions for parts (i) and (ii).

Concept / Formula Used: Total number of matrices $= k^{m \times n}$ $m \times n$ $m \times n$ k

Solution:

(i) For a matrix of order 2×3 with entries 1 or 2:

$$\begin{aligned}\text{Order of the matrix} &= 2 \times 3 \\ \text{Total number of elements in the matrix} &= 2 \times 3 = 6\end{aligned}$$

Each of the 6 positions can be independently filled by either 1 or 2.

$$\begin{aligned}\text{Number of choices for each entry} &= 2 \\ \text{Total number of possible matrices} &= 2^6 \\ &= 64\end{aligned}$$

(ii) For a matrix of order 2×2 with entries 1, 2 or 3:

$$\begin{aligned}\text{Order of the matrix} &= 2 \times 2 \\ \text{Total number of elements in the matrix} &= 2 \times 2 = 4\end{aligned}$$

Each of the 4 positions can be independently filled by 1, 2, or 3.

$$\begin{aligned}\text{Number of choices for each entry} &= 3 \\ \text{Total number of possible matrices} &= 3^4 \\ &= 81\end{aligned}$$

Final Answer

(i) Number of possible matrices $= 64$
(ii) Number of possible matrices $= 81$

Common Mistake: Confusing the base and the exponent by computing $(\text{number of elements})^{\text{number of choices}}$ instead of $(\text{number of choices})^{\text{number of elements}}$.

3.2 – Addition, Subtraction and Scalar Multiplication of Matrices

Question 1

If A and B are matrices of order $3 \times n$ and $m \times 5$ respectively, then find the order of the matrix $5A - 3B$, given that it is defined.

TYPE 1 **Matrix Order Analysis**

Given: Order of matrix $A = 3 \times n$, Order of matrix $B = m \times 5$, Matrix $5A - 3B$ is defined.

To Find: Order of matrix $5A - 3B$.

Concept / Formula Used: kA

Solution:

The order of scalar multiple matrix $5A$ is the same as the order of matrix A, which is $3 \times n$.

The order of scalar multiple matrix $3B$ is the same as the order of matrix B, which is $m \times 5$.

Since the matrix expression $5A - 3B$ is defined, matrices $5A$ and $3B$ must have identical orders.

Equating corresponding dimensions:

$$\text{Number of rows of } 5A = \text{Number of rows of } 3B \implies 3 = m$$
$$\text{Number of columns of } 5A = \text{Number of columns of } 3B \implies n = 5$$

Substituting $m = 3$ and $n = 5$, both matrices $5A$ and $3B$ have order 3×5.

The resulting matrix $5A - 3B$ therefore retains the order 3×5.

Final Answer

Order of matrix $5A - 3B = 3 \times 5$

Question 2

If $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix}$, find the following:

(i) $A + B$
(ii) $A - B$
(iii) $3A - C$

TYPE 7 **Matrix Addition Subtraction and Scalar Multiplication**

★ Likely Exam Question

Given: $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$, $C = \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix}$.

To Find: Matrices $A + B$, $A - B$, and $3A - C$.

Concept / Formula Used: $A = [a_{ij}]$ $B = [b_{ij}]$ $A \pm B = [a_{ij} \pm b_{ij}]$ $kA = [ka_{ij}]$

Solution:

(i) Evaluation of $A + B$:

$$\begin{aligned} A + B &= \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 2+1 & 4+3 \\ 3+(-2) & 2+5 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 7 \\ 1 & 7 \end{bmatrix} \end{aligned}$$

(ii) Evaluation of $A - B$:

$$\begin{aligned} A - B &= \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 2-1 & 4-3 \\ 3-(-2) & 2-5 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 5 & -3 \end{bmatrix} \end{aligned}$$

(iii) Evaluation of $3A - C$: First, perform the scalar multiplication $3A$:

$$\begin{aligned} 3A &= 3\begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3\times 2 & 3\times 4 \\ 3\times 3 & 3\times 2 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 12 \\ 9 & 6 \end{bmatrix} \end{aligned}$$

Now subtract C from $3A$:

$$\begin{aligned} 3A - C &= \begin{bmatrix} 6 & 12 \\ 9 & 6 \end{bmatrix} - \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 6-(-2) & 12-5 \\ 9-3 & 6-4 \end{bmatrix} \\ &= \begin{bmatrix} 8 & 7 \\ 6 & 2 \end{bmatrix} \end{aligned}$$

Final Answer

(i) $A + B = \begin{bmatrix} 3 & 7 \\ 1 & 7 \end{bmatrix}$

(ii) $A - B = \begin{bmatrix} 1 & 1 \\ 5 & -3 \end{bmatrix}$

(iii) $3A - C = \begin{bmatrix} 8 & 7 \\ 6 & 2 \end{bmatrix}$

Common Mistake: Watch out for double negatives during subtraction, such as $3 - (-2) = 5$ and $6 - (-2) = 8$.

Question 3

Compute the following:

(i) $\begin{bmatrix} -1 & 4 & -6 \\ 8 & 5 & 16 \\ 2 & 8 & 5 \end{bmatrix} + \begin{bmatrix} 12 & 7 & 6 \\ 8 & 0 & 5 \\ 3 & 2 & 4 \end{bmatrix}$

(ii) $\begin{bmatrix} \cos^2 x & \sin^2 x \\ \sin^2 x & \cos^2 x \end{bmatrix} + \begin{bmatrix} \sin^2 x & \cos^2 x \\ \cos^2 x & \sin^2 x \end{bmatrix}$

TYPE 7 **Matrix Addition Subtraction and Scalar Multiplication**

★ Likely Exam Question

Concept / Formula Used: $\sin^2 x + \cos^2 x = 1$

Solution:

(i) Adding corresponding elements of the 3×3 matrices:

$$\begin{bmatrix} -1 & 4 & -6 \\ 8 & 5 & 16 \\ 2 & 8 & 5 \end{bmatrix} + \begin{bmatrix} 12 & 7 & 6 \\ 8 & 0 & 5 \\ 3 & 2 & 4 \end{bmatrix}$$
$$= \begin{bmatrix} -1+12 & 4+7 & -6+6 \\ 8+8 & 5+0 & 16+5 \\ 2+3 & 8+2 & 5+4 \end{bmatrix}$$
$$= \begin{bmatrix} 11 & 11 & 0 \\ 16 & 5 & 21 \\ 5 & 10 & 9 \end{bmatrix}$$

(ii) Adding corresponding elements of the 2×2 trigonometric matrices:

$$\begin{bmatrix} \cos^2 x & \sin^2 x \\ \sin^2 x & \cos^2 x \end{bmatrix} + \begin{bmatrix} \sin^2 x & \cos^2 x \\ \cos^2 x & \sin^2 x \end{bmatrix}$$
$$= \begin{bmatrix} \cos^2 x + \sin^2 x & \sin^2 x + \cos^2 x \\ \sin^2 x + \cos^2 x & \cos^2 x + \sin^2 x \end{bmatrix}$$

Applying the standard identity $\cos^2 x + \sin^2 x = 1$ to each entry:

$$= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Final Answer

(i) $\begin{bmatrix} 11 & 11 & 0 \\ 16 & 5 & 21 \\ 5 & 10 & 9 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Question 4

If $A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 4 & -7 \end{bmatrix}$, find $2A - 3B$.

TYPE 7 **Matrix Addition Subtraction and Scalar Multiplication**

Given: $A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 4 & -7 \end{bmatrix}$.

To Find: The matrix $2A - 3B$.

Concept / Formula Used: $A, B \qquad kA + mB = [ka_{ij} + mb_{ij}]$

Solution:

First, calculate $2A$ by multiplying each entry of A by 2:

$$\begin{aligned} 2A &= 2\begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 \times 1 & 2 \times 5 \\ 2 \times (-2) & 2 \times 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 10 \\ -4 & 6 \end{bmatrix} \end{aligned}$$

Next, calculate $3B$ by multiplying each entry of B by 3:

$$\begin{aligned} 3B &= 3\begin{bmatrix} 0 & -1 \\ 4 & -7 \end{bmatrix} \\ &= \begin{bmatrix} 3 \times 0 & 3 \times (-1) \\ 3 \times 4 & 3 \times (-7) \end{bmatrix} \\ &= \begin{bmatrix} 0 & -3 \\ 12 & -21 \end{bmatrix} \end{aligned}$$

Subtract $3B$ from $2A$:

$$\begin{aligned}2A - 3B &= \begin{bmatrix} 2 & 10 \\ -4 & 6 \end{bmatrix} - \begin{bmatrix} 0 & -3 \\ 12 & -21 \end{bmatrix} \\ &= \begin{bmatrix} 2-0 & 10-(-3) \\ -4-12 & 6-(-21) \end{bmatrix} \\ &= \begin{bmatrix} 2 & 13 \\ -16 & 27 \end{bmatrix}\end{aligned}$$

Final Answer

$2A - 3B = \begin{bmatrix} 2 & 13 \\ -16 & 27 \end{bmatrix}$

Question 5

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$, then find $2A - B$.

TYPE 7 Matrix Addition Subtraction and Scalar Multiplication

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$.

To Find: The matrix $2A - B$.

Concept / Formula Used: $(2A - B)_{ij} = 2a_{ij} - b_{ij}$

Solution:

Compute $2A$ by multiplying each entry of matrix A by 2:

$$\begin{aligned}2A &= 2\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix}\end{aligned}$$

Subtract matrix B from $2A$:

$$\begin{aligned}2A - B &= \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} - \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2-3 & 4-(-1) & 6-3 \\ 4-(-1) & 6-0 & 2-2 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 4+1 & 3 \\ 4+1 & 6 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix}\end{aligned}$$

Final Answer

$$2A - B = \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix}$$

Question 6

If $\begin{bmatrix} 9 & -1 & 4 \\ -2 & 1 & 3 \end{bmatrix} = A + \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$, then find the matrix A.

TYPE 8 **Solving Linear Matrix Equations**

Given: $\begin{bmatrix} 9 & -1 & 4 \\ -2 & 1 & 3 \end{bmatrix} = A + \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$.

To Find: The matrix A.

Concept / Formula Used: $P = A + Q \qquad A = P - Q$

Solution:

Isolate A by subtracting $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$ from both sides of the matrix equation:

$$A = \begin{bmatrix} 9 & -1 & 4 \\ -2 & 1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$$

Subtract corresponding elements:

$$\begin{aligned} A &= \begin{bmatrix} 9-1 & -1-2 & 4-(-1) \\ -2-0 & 1-4 & 3-9 \end{bmatrix} \\ &= \begin{bmatrix} 8 & -3 & 4+1 \\ -2 & -3 & -6 \end{bmatrix} \\ &= \begin{bmatrix} 8 & -3 & 5 \\ -2 & -3 & -6 \end{bmatrix} \end{aligned}$$

Final Answer

$$A = \begin{bmatrix} 8 & -3 & 5 \\ -2 & -3 & -6 \end{bmatrix}$$

Question 7

If $A =$ diagonal $[1, -2, 5]$, $B =$ diagonal $[3, 0, -4]$ and $C =$ diagonal $[-2, 7, 0]$, then find $A + 2B - 3C$.

TYPE 7 **Matrix Addition Subtraction and Scalar Multiplication**

Given: $A = \text{diag}[1, -2, 5]$, $B = \text{diag}[3, 0, -4]$, $C = \text{diag}[-2, 7, 0]$.

To Find: The diagonal matrix $A + 2B - 3C$.

Concept / Formula Used: $k_1\text{diag}[d_1] + k_2\text{diag}[d_2] = \text{diag}[k_1d_1 + k_2d_2]$

Solution:

Express scalar multiples of diagonal matrices component-wise:

$$2B = 2 \cdot \text{diag}[3, 0, -4] = \text{diag}[2(3), 2(0), 2(-4)] = \text{diag}[6, 0, -8]$$
$$3C = 3 \cdot \text{diag}[-2, 7, 0] = \text{diag}[3(-2), 3(7), 3(0)] = \text{diag}[-6, 21, 0]$$

Now substitute into the expression $A + 2B - 3C$:

$$\begin{aligned} A + 2B - 3C &= \text{diag}[1, -2, 5] + \text{diag}[6, 0, -8] - \text{diag}[-6, 21, 0] \\ &= \text{diag}[1 + 6 - (-6),\ -2 + 0 - 21,\ 5 + (-8) - 0] \\ &= \text{diag}[1 + 6 + 6,\ -2 - 21,\ 5 - 8] \\ &= \text{diag}[13, -23, -3] \end{aligned}$$

Final Answer

$A + 2B - 3C = \text{diagonal } [13, -23, -3]$

Question 8

If $A = \begin{bmatrix} 2 & -3 \\ 5 & 0 \end{bmatrix}$ and $kA = \begin{bmatrix} 8 & 3a \\ -2b & c \end{bmatrix}$, find the values of k, a, b and c.

TYPE 5 **Equality of Matrices**

Given: $A = \begin{bmatrix} 2 & -3 \\ 5 & 0 \end{bmatrix}$, $kA = \begin{bmatrix} 8 & 3a \\ -2b & c \end{bmatrix}$.

To Find: The scalar values k, a, b, c.

Solution:

Compute kA by multiplying matrix A by scalar k:

$$kA = k\begin{bmatrix} 2 & -3 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 2k & -3k \\ 5k & 0 \end{bmatrix}$$

Equate kA to the given matrix:

$$\begin{bmatrix} 2k & -3k \\ 5k & 0 \end{bmatrix} = \begin{bmatrix} 8 & 3a \\ -2b & c \end{bmatrix}$$

Comparing corresponding entries step-by-step:

$$\begin{aligned} &1) \text{ Entry } (1,1): \quad 2k = 8 \implies k = \frac{8}{2} \implies k = 4 \\ &2) \text{ Entry } (1,2): \quad -3k = 3a \implies -3(4) = 3a \implies -12 = 3a \implies a = -4 \\ &3) \text{ Entry } (2,1): \quad 5k = -2b \implies 5(4) = -2b \implies 20 = -2b \implies b = -10 \\ &4) \text{ Entry } (2,2): \quad 0 = c \implies c = 0 \end{aligned}$$

Final Answer

$k = 4$

$a = -4$

$b = -10$

$c = 0$

Question 9

If $x\begin{bmatrix}2\\3\end{bmatrix} + y\begin{bmatrix}-1\\1\end{bmatrix} = \begin{bmatrix}10\\5\end{bmatrix}$, find the values of x and y.

TYPE 5 **Equality of Matrices**

Given: $x\begin{bmatrix}2\\3\end{bmatrix} + y\begin{bmatrix}-1\\1\end{bmatrix} = \begin{bmatrix}10\\5\end{bmatrix}$.

To Find: The values of x and y.

Concept / Formula Used: $xC_1 + yC_2 = B$ x y

Solution:

Perform scalar multiplication on the column vectors:

$$\begin{bmatrix}2x\\3x\end{bmatrix} + \begin{bmatrix}-y\\y\end{bmatrix} = \begin{bmatrix}10\\5\end{bmatrix}$$

Combine LHS into a single matrix:

$$\begin{bmatrix}2x-y\\3x+y\end{bmatrix} = \begin{bmatrix}10\\5\end{bmatrix}$$

Equating corresponding entries yields two simultaneous linear equations:

$$2x - y = 10 \tag{1}$$
$$3x + y = 5 \tag{2}$$

Add equation (1) and equation (2) to eliminate y:

$$\begin{aligned}(2x-y)+(3x+y) &= 10+5\\ 5x &= 15\\ x &= \frac{15}{5}\\ x &= 3\end{aligned}$$

Substitute $x = 3$ into equation (2):

$$\begin{aligned}3(3)+y &= 5\\ 9+y &= 5\\ y &= 5-9\\ y &= -4\end{aligned}$$

Final Answer

$x = 3, \quad y = -4$

Question 10

If $2\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} -3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$, find the value of $(x - y)$.

TYPE 5 **Equality of Matrices**

Given: $2\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} -3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$.

To Find: The value of $(x - y)$.

Solution:

Multiply scalar 2 inside the matrix:

$$\begin{bmatrix} 2x & 10 \\ 14 & 2(y-3) \end{bmatrix} + \begin{bmatrix} -3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

Expand $2(y - 3) = 2y - 6$ and perform matrix addition on LHS:

$$\begin{bmatrix} 2x-3 & 10-4 \\ 14+1 & 2y-6+2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 2x-3 & 6 \\ 15 & 2y-4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

Equate corresponding elements:

$$2x - 3 = 7 \implies 2x = 10 \implies x = 5$$
$$2y - 4 = 14 \implies 2y = 18 \implies y = 9$$

Now compute $x - y$:

$$\begin{aligned} x - y &= 5 - 9 \\ &= -4 \end{aligned}$$

Final Answer

$x - y = -4$

Question 11

Given $2\begin{bmatrix} x & z \\ y & t \end{bmatrix} + 3\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3\begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$, find x, y, z, t.

TYPE 5 **Equality of Matrices**

★ **Likely Exam Question**

Given: $2\begin{bmatrix} x & z \\ y & t \end{bmatrix} + 3\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3\begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$.

To Find: The values of x, y, z, t.

Solution:

Multiply scalars into each matrix:

$$\begin{bmatrix} 2x & 2z \\ 2y & 2t \end{bmatrix} + \begin{bmatrix} 3 & -3 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 12 & 18 \end{bmatrix}$$

Add the two matrices on the left-hand side:

$$\begin{bmatrix} 2x+3 & 2z-3 \\ 2y+0 & 2t+6 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 12 & 18 \end{bmatrix}$$

Equate corresponding entries individually:

$$1) \text{ Entry } (1,1): \quad 2x+3=9 \implies 2x=6 \implies x=3$$
$$2) \text{ Entry } (1,2): \quad 2z-3=15 \implies 2z=18 \implies z=9$$
$$3) \text{ Entry } (2,1): \quad 2y=12 \implies y=6$$
$$4) \text{ Entry } (2,2): \quad 2t+6=18 \implies 2t=12 \implies t=6$$

Final Answer

$x=3, \quad y=6, \quad z=9, \quad t=6$

Question 12

If $3\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 6 \\ -1 & 2d \end{bmatrix} + \begin{bmatrix} 4 & a+b \\ c+d & 3 \end{bmatrix}$, find a, b, c and d.

TYPE 5 **Equality of Matrices**

★ **Likely Exam Question**

Given: $3\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 6 \\ -1 & 2d \end{bmatrix} + \begin{bmatrix} 4 & a+b \\ c+d & 3 \end{bmatrix}$.

To Find: The values of a, b, c, d.

Solution:

Simplify LHS and RHS matrices:

$$\begin{bmatrix} 3a & 3b \\ 3c & 3d \end{bmatrix} = \begin{bmatrix} a+4 & 6+a+b \\ -1+c+d & 2d+3 \end{bmatrix}$$

Equating corresponding entries step-by-step:

1) From entry $(1, 1)$:

$$\begin{aligned} 3a &= a + 4 \\ 3a - a &= 4 \\ 2a &= 4 \\ a &= 2 \end{aligned}$$

2) From entry $(1, 2)$:

$$\begin{aligned} 3b &= 6 + a + b \\ 3b - b &= 6 + a \\ 2b &= 6 + 2 \quad \text{(substituting } a = 2\text{)} \\ 2b &= 8 \\ b &= 4 \end{aligned}$$

3) From entry $(2, 2)$:

$$\begin{aligned} 3d &= 2d + 3 \\ 3d - 2d &= 3 \\ d &= 3 \end{aligned}$$

4) From entry $(2, 1)$:

$$\begin{aligned} 3c &= -1 + c + d \\ 3c - c &= -1 + d \\ 2c &= -1 + 3 \quad \text{(substituting } d = 3\text{)} \\ 2c &= 2 \\ c &= 1 \end{aligned}$$

Final Answer

$a = 2, \quad b = 4, \quad c = 1, \quad d = 3$

Question 13

Find X if $Y = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $2X + Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$.

TYPE 8 Solving Linear Matrix Equations

★ Likely Exam Question

Given: $Y = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, $2X + Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$.

To Find: The matrix X.

Concept / Formula Used: $2X + Y = B \qquad 2X = B - Y \implies X = \frac{1}{2}(B - Y)$

Solution:

Substitute matrix Y into the equation $2X + Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$:

$$2X + \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$$

Isolate $2X$ by subtracting $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ from both sides:

$$\begin{aligned} 2X &= \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 1-3 & 0-2 \\ -3-1 & 2-4 \end{bmatrix} \\ &= \begin{bmatrix} -2 & -2 \\ -4 & -2 \end{bmatrix} \end{aligned}$$

Multiply by $\frac{1}{2}$ to solve for X:

$$\begin{aligned} X &= \frac{1}{2}\begin{bmatrix} -2 & -2 \\ -4 & -2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{-2}{2} & \frac{-2}{2} \\ \frac{-4}{2} & \frac{-2}{2} \end{bmatrix} \\ &= \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix} \end{aligned}$$

Final Answer

$X = \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix}$

Question 14

Find matrices X and Y, if $X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$ and $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

TYPE 8 **Solving Linear Matrix Equations**

Given: $X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$, $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

To Find: The matrices X and Y.

Concept / Formula Used: $X + Y = A \qquad X - Y = B \qquad 2X = A + B \qquad 2Y = A - B$

Solution:

Given simultaneous matrix equations:

$$X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} \tag{1}$$

$$X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \tag{2}$$

Add equation (1) and equation (2) to eliminate Y:

$$\begin{aligned}
(X + Y) + (X - Y) &= \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\
2X &= \begin{bmatrix} 7+3 & 0+0 \\ 2+0 & 5+3 \end{bmatrix} \\
2X &= \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} \\
X &= \frac{1}{2}\begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} \\
X &= \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}
\end{aligned}$$

Subtract equation (2) from equation (1) to eliminate X:

$$\begin{aligned}
(X + Y) - (X - Y) &= \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\
2Y &= \begin{bmatrix} 7-3 & 0-0 \\ 2-0 & 5-3 \end{bmatrix} \\
2Y &= \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} \\
Y &= \frac{1}{2}\begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} \\
Y &= \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}
\end{aligned}$$

Final Answer

$X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$

$Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

Question 15

Find matrices X and Y if $2X - Y = \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix}$ and $X + 2Y = \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix}$.

TYPE 8 **Solving Linear Matrix Equations**

Given: $2X - Y = \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix}$, $X + 2Y = \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix}$.

To Find: The matrices X and Y.

Solution:

Label the given matrix equations:

$$2X - Y = \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix} \tag{1}$$

$$X + 2Y = \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix} \tag{2}$$

Multiply equation (1) by 2:

$$4X - 2Y = 2\begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 12 & -12 & 0 \\ -8 & 4 & 2 \end{bmatrix} \tag{3}$$

Add equation (2) and equation (3) to eliminate Y:

$$\begin{aligned}
(X + 2Y) + (4X - 2Y) &= \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix} + \begin{bmatrix} 12 & -12 & 0 \\ -8 & 4 & 2 \end{bmatrix} \\
5X &= \begin{bmatrix} 3+12 & 2+(-12) & 5+0 \\ -2+(-8) & 1+4 & -7+2 \end{bmatrix} \\
5X &= \begin{bmatrix} 15 & -10 & 5 \\ -10 & 5 & -5 \end{bmatrix} \\
X &= \frac{1}{5}\begin{bmatrix} 15 & -10 & 5 \\ -10 & 5 & -5 \end{bmatrix} \\
X &= \begin{bmatrix} 3 & -2 & 1 \\ -2 & 1 & -1 \end{bmatrix}
\end{aligned}$$

Multiply equation (2) by 2:

$$2X + 4Y = 2\begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix} = \begin{bmatrix} 6 & 4 & 10 \\ -4 & 2 & -14 \end{bmatrix} \tag{4}$$

Subtract equation (1) from equation (4) to eliminate X:

$$\begin{aligned}(2X+4Y)-(2X-Y) &= \begin{bmatrix}6 & 4 & 10\\ -4 & 2 & -14\end{bmatrix} - \begin{bmatrix}6 & -6 & 0\\ -4 & 2 & 1\end{bmatrix}\\ 5Y &= \begin{bmatrix}6-6 & 4-(-6) & 10-0\\ -4-(-4) & 2-2 & -14-1\end{bmatrix}\\ 5Y &= \begin{bmatrix}0 & 10 & 10\\ 0 & 0 & -15\end{bmatrix}\\ Y &= \frac{1}{5}\begin{bmatrix}0 & 10 & 10\\ 0 & 0 & -15\end{bmatrix}\\ Y &= \begin{bmatrix}0 & 2 & 2\\ 0 & 0 & -3\end{bmatrix}\end{aligned}$$

Final Answer

$X = \begin{bmatrix}3 & -2 & 1\\ -2 & 1 & -1\end{bmatrix}$

$Y = \begin{bmatrix}0 & 2 & 2\\ 0 & 0 & -3\end{bmatrix}$

Question 16

Find a matrix X such that $3A - 2B + X = O$, where $A = \begin{bmatrix}4 & 2\\ 1 & 3\end{bmatrix}$, $B = \begin{bmatrix}-2 & 1\\ 3 & 2\end{bmatrix}$.

TYPE 8 Solving Linear Matrix Equations

Given: $A = \begin{bmatrix}4 & 2\\ 1 & 3\end{bmatrix}$, $B = \begin{bmatrix}-2 & 1\\ 3 & 2\end{bmatrix}$, $3A - 2B + X = O$.

To Find: The matrix X.

Concept / Formula Used: $3A - 2B + X = O \qquad X = 2B - 3A$

Solution:

Rearrange the matrix equation to isolate X:

$$\begin{aligned}3A - 2B + X &= O\\ X &= 2B - 3A\end{aligned}$$

Calculate $2B$ by scalar multiplication:

$$\begin{aligned}2B &= 2\begin{bmatrix}-2 & 1\\ 3 & 2\end{bmatrix}\\ &= \begin{bmatrix}-4 & 2\\ 6 & 4\end{bmatrix}\end{aligned}$$

Calculate $3A$ by scalar multiplication:

$$\begin{aligned} 3A &= 3\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 12 & 6 \\ 3 & 9 \end{bmatrix} \end{aligned}$$

Subtract $3A$ from $2B$:

$$\begin{aligned} X &= \begin{bmatrix} -4 & 2 \\ 6 & 4 \end{bmatrix} - \begin{bmatrix} 12 & 6 \\ 3 & 9 \end{bmatrix} \\ &= \begin{bmatrix} -4-12 & 2-6 \\ 6-3 & 4-9 \end{bmatrix} \\ &= \begin{bmatrix} -16 & -4 \\ 3 & -5 \end{bmatrix} \end{aligned}$$

Final Answer

$X = \begin{bmatrix} -16 & -4 \\ 3 & -5 \end{bmatrix}$

Question 17

If $A = \begin{bmatrix} 9 & 1 \\ 7 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix}$, find matrix C such that $5A + 3B + 2C$ is a null matrix.

TYPE 8 Solving Linear Matrix Equations

★ **Likely Exam Question**

Given: $A = \begin{bmatrix} 9 & 1 \\ 7 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix}$, $5A + 3B + 2C = O$.

To Find: The matrix C.

Concept / Formula Used: O $\quad 5A + 3B + 2C = O \quad C = -\frac{1}{2}(5A + 3B)$

Solution:

Rearrange the equation to express $2C$:

$$\begin{aligned} 5A + 3B + 2C &= O \\ 2C &= -(5A + 3B) \\ C &= -\frac{1}{2}(5A + 3B) \end{aligned}$$

Compute $5A$:

$$\begin{aligned} 5A &= 5\begin{bmatrix} 9 & 1 \\ 7 & 8 \end{bmatrix} \\ &= \begin{bmatrix} 45 & 5 \\ 35 & 40 \end{bmatrix} \end{aligned}$$

Compute $3B$:

$$\begin{aligned} 3B &= 3\begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 15 \\ 21 & 36 \end{bmatrix} \end{aligned}$$

Add $5A$ and $3B$:

$$\begin{aligned} 5A + 3B &= \begin{bmatrix} 45 & 5 \\ 35 & 40 \end{bmatrix} + \begin{bmatrix} 3 & 15 \\ 21 & 36 \end{bmatrix} \\ &= \begin{bmatrix} 45+3 & 5+15 \\ 35+21 & 40+36 \end{bmatrix} \\ &= \begin{bmatrix} 48 & 20 \\ 56 & 76 \end{bmatrix} \end{aligned}$$

Multiply by $-\frac{1}{2}$ to get matrix C:

$$\begin{aligned} C &= -\frac{1}{2}\begin{bmatrix} 48 & 20 \\ 56 & 76 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{48}{2} & -\frac{20}{2} \\ -\frac{56}{2} & -\frac{76}{2} \end{bmatrix} \\ &= \begin{bmatrix} -24 & -10 \\ -28 & -38 \end{bmatrix} \end{aligned}$$

Final Answer

$C = \begin{bmatrix} -24 & -10 \\ -28 & -38 \end{bmatrix}$

3.3 – Multiplication of Matrices

Question 1

If $A = [-1 \quad 2 \quad -5]$ and $B = \begin{bmatrix} 2 \\ -1 \\ 7 \end{bmatrix}$, write the orders of AB and BA.

TYPE 9 Matrix Multiplication Compatibility and Order

Given: Matrix A of order 1×3 and matrix B of order 3×1

To Find: The orders of the matrix products AB and BA

Concept / Formula Used: $P \quad m \times n \quad Q \quad n \times p \quad PQ \quad m \times p$

Solution:

The order of matrix A is 1×3.
The order of matrix B is 3×1.

For the product AB:

$$\text{Order of } A = 1 \times 3$$
$$\text{Order of } B = 3 \times 1$$

Using the matrix multiplication compatibility rule, since the number of columns in A (3) equals the number of rows in B (3), AB is defined and its order is:

$$\text{Order of } AB = 1 \times 1$$

For the product BA:

$$\text{Order of } B = 3 \times 1$$
$$\text{Order of } A = 1 \times 3$$

Since the number of columns in B (1) equals the number of rows in A (1), BA is defined and its order is:

$$\text{Order of } BA = 3 \times 3$$

Final Answer

The order of AB is 1×1 and the order of BA is 3×3.

Question 2

If $A = \begin{bmatrix} 2 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$, then write the orders of AB and BA.

TYPE 9 Matrix Multiplication Compatibility and Order

Given: Matrix $A = \begin{bmatrix} 2 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$

To Find: The orders of the products AB and BA

Concept / Formula Used: $m \times n$ $n \times p$ $m \times p$

Solution:

By counting rows and columns:

$$\text{Order of } A = 2 \times 3$$
$$\text{Order of } B = 3 \times 2$$

For AB:

$$\text{Number of columns of } A = 3$$
$$\text{Number of rows of } B = 3$$

Since these are equal, AB exists and its order is given by:

$$\text{Order of } AB = 2 \times 2$$

For BA:

$$\text{Number of columns of } B = 2$$
$$\text{Number of rows of } A = 2$$

Since these are equal, BA exists and its order is given by:

$$\text{Order of } BA = 3 \times 3$$

Final Answer

The order of AB is 2×2 and the order of BA is 3×3.

Question 3

If A is a matrix of order 3×3 and B is a matrix of order 2×3, can you find AB?

TYPE 9 **Matrix Multiplication Compatibility and Order**

Given: Matrix A of order 3×3 and matrix B of order 2×3

To Find: Whether the matrix product AB can be found

Concept / Formula Used: PQ P Q

Solution:

Let us inspect the dimensions of A and B:

$$\text{Number of columns of } A = 3$$
$$\text{Number of rows of } B = 2$$

Comparing these two values:

$$3 \neq 2$$

Since the number of columns of A is not equal to the number of rows of B, the condition for matrix multiplication is not satisfied.

Final Answer

No, AB cannot be found because the number of columns in A (3) is not equal to the number of rows in B (2).

Question 4

If $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 4 & 2 \end{bmatrix}$ and $AB = [c_{ij}]$, find $c_{23} + c_{31}$.

TYPE 10 **Evaluating Specific Product Matrix Elements**

Given: $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 4 & 2 \end{bmatrix}$, $AB = [c_{ij}]$

To Find: The value of $c_{23} + c_{31}$

Concept / Formula Used: c_{ij} $C = AB$ i A j B

Solution:

The element c_{23} is the product of row 2 of A and column 3 of B:

$$\begin{aligned} \text{Row 2 of } A &= \begin{bmatrix} -1 & 1 \end{bmatrix} \\ \text{Column 3 of } B &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ c_{23} &= (-1)(1) + (1)(2) \\ c_{23} &= -1 + 2 \\ c_{23} &= 1 \end{aligned}$$

The element c_{31} is the product of row 3 of A and column 1 of B:

$$\begin{aligned} \text{Row 3 of } A &= \begin{bmatrix} 2 & 3 \end{bmatrix} \\ \text{Column 1 of } B &= \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ c_{31} &= (2)(3) + (3)(1) \\ c_{31} &= 6 + 3 \\ c_{31} &= 9 \end{aligned}$$

Adding the two calculated entries:

$$\begin{aligned} c_{23} + c_{31} &= 1 + 9 \\ c_{23} + c_{31} &= 10 \end{aligned}$$

Final Answer

$c_{23} + c_{31} = 10$

Question 5

If $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \\ -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & 4 \end{bmatrix}$ and $BA = [c_{ij}]$, find $c_{21} + c_{22}$.

TYPE 10 **Evaluating Specific Product Matrix Elements**

Given: $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \\ -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & 4 \end{bmatrix}$, $BA = [c_{ij}]$

To Find: The value of $c_{21} + c_{22}$

Concept / Formula Used: c_{ij} BA i B j A

Solution:

To find c_{21}, multiply Row 2 of B with Column 1 of A:

$$\text{Row 2 of } B = \begin{bmatrix} 2 & 1 & 4 \end{bmatrix}$$

$$\text{Column 1 of } A = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

$$c_{21} = (2)(2) + (1)(3) + (4)(-1)$$
$$c_{21} = 4 + 3 - 4$$
$$c_{21} = 3$$

To find c_{22}, multiply Row 2 of B with Column 2 of A:

$$\text{Column 2 of } A = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$c_{22} = (2)(4) + (1)(1) + (4)(2)$$
$$c_{22} = 8 + 1 + 8$$
$$c_{22} = 17$$

Summing the two components:

$$c_{21} + c_{22} = 3 + 17$$
$$c_{21} + c_{22} = 20$$

Final Answer

$c_{21} + c_{22} = 20$

Question 6

If $\begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} = O$, find the value of x.

TYPE 11 **Solving Matrix Equations using Multiplication**

Given: $\begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$

To Find: The value of x

Solution:

Perform the matrix multiplication on the left hand side:

$$\begin{aligned} \begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} &= \begin{bmatrix} x(1) + 1(-2) & x(0) + 1(0) \end{bmatrix} \\ &= \begin{bmatrix} x - 2 & 0 \end{bmatrix} \end{aligned}$$

Equating the product to the zero matrix on the right hand side:

$$\begin{bmatrix} x - 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

Equating corresponding entries:

$$\begin{aligned} x - 2 &= 0 \\ x &= 2 \end{aligned}$$

Final Answer

$x = 2$

Question 7

If $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$, write the value of x.

★ **Likely Exam Question**

TYPE 11 **Solving Matrix Equations using Multiplication**

Given: $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$

To Find: The value of x

Concept / Formula Used: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$

Solution:

Compute the left hand side matrix product:

$$\begin{aligned} \text{LHS} &= \begin{bmatrix} 2(1)+3(-2) & 2(-3)+3(4) \\ 5(1)+7(-2) & 5(-3)+7(4) \end{bmatrix} \\ &= \begin{bmatrix} 2-6 & -6+12 \\ 5-14 & -15+28 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 6 \\ -9 & 13 \end{bmatrix} \end{aligned}$$

Given that LHS = RHS:

$$\begin{bmatrix} -4 & 6 \\ -9 & 13 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$$

Comparing the entry at position $(2, 2)$ on both sides:

$$x = 13$$

Final Answer

$x = 13$

Question 8

Write the value of $x + y + z$ if $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$.

TYPE 11 **Solving Matrix Equations using Multiplication**

Given: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

To Find: The value of $x + y + z$

Concept / Formula Used: V $\quad I \cdot V = V \quad$ I

Solution:

Using the property $IX = X$ where $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 1(x)+0(y)+0(z) \\ 0(x)+1(y)+0(z) \\ 0(x)+0(y)+1(z) \end{bmatrix} \\ &= \begin{bmatrix} x \\ y \\ z \end{bmatrix} \end{aligned}$$

Equating to the given column vector:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Comparing elements:

$$\begin{aligned} x &= 1 \\ y &= -1 \\ z &= 0 \end{aligned}$$

Now compute $x + y + z$:

$$\begin{aligned} x + y + z &= 1 + (-1) + 0 \\ &= 0 \end{aligned}$$

Final Answer

$x + y + z = 0$

Question 9

Compute the indicated products:

(i) $\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix}$

TYPE 12 **Matrix Multiplication**

Given: Matrix expressions for parts (i) and (ii)

To Find: The product matrix for each part

Concept / Formula Used: $(i, j) = \sum_k a_{ik} b_{kj}$

Solution:

Part (i): Let $A = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$ of order 2×2 and $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ of order 2×3.

The product AB will have order 2×3:

$$\begin{aligned} AB &= \begin{bmatrix} 1(1) + (-2)(2) & 1(2) + (-2)(3) & 1(3) + (-2)(1) \\ 2(1) + 3(2) & 2(2) + 3(3) & 2(3) + 3(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 - 4 & 2 - 6 & 3 - 2 \\ 2 + 6 & 4 + 9 & 6 + 3 \end{bmatrix} \\ &= \begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix} \end{aligned}$$

Part (ii): Let $C = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$ of order 2×3 and $D = \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix}$ of order 3×2.

The product CD will have order 2×2:

$$\begin{aligned} CD &= \begin{bmatrix} 3(2) + (-1)(1) + 3(3) & 3(-3) + (-1)(0) + 3(1) \\ (-1)(2) + 0(1) + 2(3) & (-1)(-3) + 0(0) + 2(1) \end{bmatrix} \\ &= \begin{bmatrix} 6 - 1 + 9 & -9 + 0 + 3 \\ -2 + 0 + 6 & 3 + 0 + 2 \end{bmatrix} \\ &= \begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix} \end{aligned}$$

Final Answer

(i) $\begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix}$

(ii) $\begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix}$

Question 10

If $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ and $A^2 = I$, find the value of $\alpha^2 + \beta\gamma$.

TYPE 11 **Solving Matrix Equations using Multiplication**

Given: $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ and $A^2 = I$

To Find: The value of $\alpha^2 + \beta\gamma$

Concept / Formula Used: $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ 2×2

Solution:

First calculate $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix} \\ &= \begin{bmatrix} \alpha(\alpha) + \beta(\gamma) & \alpha(\beta) + \beta(-\alpha) \\ \gamma(\alpha) + (-\alpha)(\gamma) & \gamma(\beta) + (-\alpha)(-\alpha) \end{bmatrix} \\ &= \begin{bmatrix} \alpha^2 + \beta\gamma & \alpha\beta - \alpha\beta \\ \alpha\gamma - \alpha\gamma & \beta\gamma + \alpha^2 \end{bmatrix} \\ &= \begin{bmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \alpha^2 + \beta\gamma \end{bmatrix} \end{aligned}$$

Given $A^2 = I$:

$$\begin{bmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \alpha^2 + \beta\gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Comparing the entry at row 1, column 1:

$$\alpha^2 + \beta\gamma = 1$$

Final Answer

$\alpha^2 + \beta\gamma = 1$

Question 11

If $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find AB and BA.

TYPE 12 **Matrix Multiplication**

Given: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

To Find: The matrices AB and BA

Concept / Formula Used: 2×2

Solution:

Calculating AB:

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1(0) + 0(1) & 1(1) + 0(0) \\ 0(0) + (-1)(1) & 0(1) + (-1)(0) \end{bmatrix} \\ &= \begin{bmatrix} 0 + 0 & 1 + 0 \\ 0 - 1 & 0 + 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \end{aligned}$$

Calculating BA:

$$\begin{aligned} BA &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 0(1) + 1(0) & 0(0) + 1(-1) \\ 1(1) + 0(0) & 1(0) + 0(-1) \end{bmatrix} \\ &= \begin{bmatrix} 0 + 0 & 0 - 1 \\ 1 + 0 & 0 + 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

Final Answer

$AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Question 12

(i) If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, find AB and BA.

(ii) If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ and $AB = BA$, then show that B is a scalar matrix.

TYPE 13 **Matrix Multiplication Properties and Proofs**

Given: (i) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
(ii) $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, $AB = BA$

To Prove: (i) Find AB and BA
(ii) Show that B is a scalar matrix (i.e. $a = b$)

Concept / Formula Used: $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ $\qquad a = b$

Solution:

Part (i):

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1(1)+1(-1) & 1(-1)+1(1) \\ 1(1)+1(-1) & 1(-1)+1(1) \end{bmatrix} \\ &= \begin{bmatrix} 1-1 & -1+1 \\ 1-1 & -1+1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \end{aligned}$$

$$\begin{aligned}
BA &= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1(1) + (-1)(1) & 1(1) + (-1)(1) \\ (-1)(1) + 1(1) & (-1)(1) + 1(1) \end{bmatrix} \\
&= \begin{bmatrix} 1 - 1 & 1 - 1 \\ -1 + 1 & -1 + 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O
\end{aligned}$$

Part (ii): Calculate product AB:

$$\begin{aligned}
AB &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \\
&= \begin{bmatrix} 1(a) + 2(0) & 1(0) + 2(b) \\ 3(a) + 4(0) & 3(0) + 4(b) \end{bmatrix} \\
&= \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix}
\end{aligned}$$

Calculate product BA:

$$\begin{aligned}
BA &= \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\
&= \begin{bmatrix} a(1) + 0(3) & a(2) + 0(4) \\ 0(1) + b(3) & 0(2) + b(4) \end{bmatrix} \\
&= \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}
\end{aligned}$$

Since $AB = BA$:

$$\begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix} = \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}$$

Equating corresponding entry at $(1, 2)$:

$$\begin{aligned}
2b &= 2a \\
a &= b
\end{aligned}$$

Similarly, equating entry at $(2, 1)$:

$$\begin{aligned}
3a &= 3b \\
a &= b
\end{aligned}$$

Since $a = b$, matrix B becomes $B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$, which is by definition a scalar matrix.

Hence Proved

Part (i) yields $AB = O$ and $BA = O$. Part (ii) shows $a = b$, hence B is a scalar matrix.

Question 13

If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ and $A^2 = B$, then find the value(s) of α.

TYPE 11 **Solving Matrix Equations using Multiplication**

Given: $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, and $A^2 = B$

To Find: The value(s) of α

Solution:

Compute $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \alpha(\alpha) + 0(1) & \alpha(0) + 0(1) \\ 1(\alpha) + 1(1) & 1(0) + 1(1) \end{bmatrix} \\ &= \begin{bmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{bmatrix} \end{aligned}$$

Set $A^2 = B$:

$$\begin{bmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

Equating corresponding elements gives two conditions on α:

$$\alpha^2 = 1 \implies \alpha = 1 \text{ or } \alpha = -1 \tag{1}$$

$$\alpha + 1 = 2 \implies \alpha = 1 \tag{2}$$

For $A^2 = B$ to hold, α must satisfy both equations simultaneously. The common value satisfying both equations is:

$$\alpha = 1$$

Final Answer

$\alpha = 1$

Question 14

If A is a square matrix such that $A^2 = I$, then find the simplified value of $(A - I)^3 + (A + I)^3 - 7A$.

★ **Likely Exam Question**

TYPE 14 **Simplifying Matrix Polynomial Expressions**

Given: A is a square matrix satisfying $A^2 = I$

To Find: The simplified expression for $(A-I)^3+(A+I)^3-7A$

Concept / Formula Used: A I $AI = IA = A$ $(A-I)^3 = A^3 - 3A^2I + 3AI^2 - I^3$ $(A+I)^3 = A^3 + 3A^2I + 3AI^2 + I^3$

Solution:

Given $A^2 = I$, we first compute higher powers of A:

$$\begin{aligned} A^3 &= A^2 \cdot A \\ &= I \cdot A \\ &= A \end{aligned}$$

Also note the properties of identity matrix powers:

$$\begin{aligned} I^2 &= I \\ I^3 &= I \\ A^2I &= I \cdot I = I \\ AI^2 &= A \cdot I = A \end{aligned}$$

Now expand $(A-I)^3$:

$$\begin{aligned} (A-I)^3 &= A^3 - 3A^2I + 3AI^2 - I^3 \\ &= A - 3I + 3A - I \\ &= 4A - 4I \end{aligned}$$

Next expand $(A+I)^3$:

$$\begin{aligned} (A+I)^3 &= A^3 + 3A^2I + 3AI^2 + I^3 \\ &= A + 3I + 3A + I \\ &= 4A + 4I \end{aligned}$$

Now substitute both expansions into the target expression:

$$\begin{aligned} (A-I)^3 + (A+I)^3 - 7A &= (4A - 4I) + (4A + 4I) - 7A \\ &= 4A + 4A - 7A - 4I + 4I \\ &= 8A - 7A + O \\ &= A \end{aligned}$$

Final Answer

$(A-I)^3 + (A+I)^3 - 7A = A$

Question 15

Write the following as a single matrix:

(i) $\begin{bmatrix} 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 5 \\ 0 & 2 & 4 \\ 7 & 5 & 0 \end{bmatrix} - \begin{bmatrix} 2 & -5 & 7 \end{bmatrix}$

(ii) $\begin{bmatrix} 3 & 2 & 5 \\ 7 & -4 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & -1 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 7 & -8 \\ 5 & 9 \end{bmatrix}$

TYPE 15 **Matrix Operations and Simplification**

Given: Matrix expressions combining multiplication and subtraction for parts (i) and (ii)

To Find: Single simplified matrix for each part

Solution:

Part (i): First perform the product of the row matrix 1×3 and 3×3 matrix:

$$\begin{aligned}
\begin{bmatrix} 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 5 \\ 0 & 2 & 4 \\ 7 & 5 & 0 \end{bmatrix} &= \begin{bmatrix} 1(2)+(-2)(0)+3(7) & 1(-1)+(-2)(2)+3(5) & 1(5)+(-2)(4)+3(0) \end{bmatrix} \\
&= \begin{bmatrix} 2+0+21 & -1-4+15 & 5-8+0 \end{bmatrix} \\
&= \begin{bmatrix} 23 & 10 & -3 \end{bmatrix}
\end{aligned}$$

Now perform matrix subtraction:

$$\begin{aligned}
\begin{bmatrix} 23 & 10 & -3 \end{bmatrix} - \begin{bmatrix} 2 & -5 & 7 \end{bmatrix} &= \begin{bmatrix} 23-2 & 10-(-5) & -3-7 \end{bmatrix} \\
&= \begin{bmatrix} 21 & 15 & -10 \end{bmatrix}
\end{aligned}$$

Part (ii): First evaluate the product of the 2×3 matrix and 3×2 matrix:

$$\begin{aligned}
\begin{bmatrix} 3 & 2 & 5 \\ 7 & -4 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & -1 \\ 3 & 5 \end{bmatrix} &= \begin{bmatrix} 3(2)+2(2)+5(3) & 3(2)+2(-1)+5(5) \\ 7(2)+(-4)(2)+0(3) & 7(2)+(-4)(-1)+0(5) \end{bmatrix} \\
&= \begin{bmatrix} 6+4+15 & 6-2+25 \\ 14-8+0 & 14+4+0 \end{bmatrix} \\
&= \begin{bmatrix} 25 & 29 \\ 6 & 18 \end{bmatrix}
\end{aligned}$$

Now perform matrix subtraction:

$$\begin{aligned}
\begin{bmatrix} 25 & 29 \\ 6 & 18 \end{bmatrix} - \begin{bmatrix} 7 & -8 \\ 5 & 9 \end{bmatrix} &= \begin{bmatrix} 25-7 & 29-(-8) \\ 6-5 & 18-9 \end{bmatrix} \\
&= \begin{bmatrix} 18 & 37 \\ 1 & 9 \end{bmatrix}
\end{aligned}$$

Final Answer

(i) $\begin{bmatrix} 21 & 15 & -10 \end{bmatrix}$

(ii) $\begin{bmatrix} 18 & 37 \\ 1 & 9 \end{bmatrix}$

Question 16

(i) If $A = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$, find A^2.

(ii) If $M(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, show that $M(x)M(y) = M(x+y)$.

TYPE 16 Matrix Multiplication with Trigonometric Identities

Given: (i) $A = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$

(ii) $M(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

To Prove: (i) Find A^2

(ii) Show that $M(x)M(y) = M(x+y)$

Concept / Formula Used: $Trigonometric addition formulae$:

$\cos A \cos B - \sin A \sin B = \cos(A+B)$

$\sin A \cos B + \cos A \sin B = \sin(A+B)$

Solution:

Part (i):

$$A^2 = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix} \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$$
$$= \begin{bmatrix} \cos^2 2\alpha - \sin^2 2\alpha & \cos 2\alpha \sin 2\alpha + \sin 2\alpha \cos 2\alpha \\ -\sin 2\alpha \cos 2\alpha - \cos 2\alpha \sin 2\alpha & -\sin^2 2\alpha + \cos^2 2\alpha \end{bmatrix}$$

Using double-angle trigonometric identities:

$$\cos^2 2\alpha - \sin^2 2\alpha = \cos(2\alpha + 2\alpha) = \cos 4\alpha$$
$$\sin 2\alpha \cos 2\alpha + \cos 2\alpha \sin 2\alpha = \sin(2\alpha + 2\alpha) = \sin 4\alpha$$

Substituting these identities back gives:

$$A^2 = \begin{bmatrix} \cos 4\alpha & \sin 4\alpha \\ -\sin 4\alpha & \cos 4\alpha \end{bmatrix}$$

Part (ii): Write down $M(x)$ and $M(y)$:

$$M(x) = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$
$$M(y) = \begin{bmatrix} \cos y & \sin y \\ -\sin y & \cos y \end{bmatrix}$$

Multiplying $M(x)$ and $M(y)$:

$$\begin{aligned} M(x)M(y) &= \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} \begin{bmatrix} \cos y & \sin y \\ -\sin y & \cos y \end{bmatrix} \\ &= \begin{bmatrix} \cos x \cos y - \sin x \sin y & \cos x \sin y + \sin x \cos y \\ -\sin x \cos y - \cos x \sin y & -\sin x \sin y + \cos x \cos y \end{bmatrix} \end{aligned}$$

Using standard trigonometric angle-addition identities:

$$\begin{aligned} \cos x \cos y - \sin x \sin y &= \cos(x+y) \\ \sin x \cos y + \cos x \sin y &= \sin(x+y) \\ -(\sin x \cos y + \cos x \sin y) &= -\sin(x+y) \\ \cos x \cos y - \sin x \sin y &= \cos(x+y) \end{aligned}$$

Substitute these into the product matrix:

$$\begin{aligned} M(x)M(y) &= \begin{bmatrix} \cos(x+y) & \sin(x+y) \\ -\sin(x+y) & \cos(x+y) \end{bmatrix} \\ &= M(x+y) \end{aligned}$$

Hence Proved

$A^2 = \begin{bmatrix} \cos 4\alpha & \sin 4\alpha \\ -\sin 4\alpha & \cos 4\alpha \end{bmatrix}$ and $M(x)M(y) = M(x+y)$ is proved.

Question 17

If $A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix}$, find $3A^2 - 2B + I$.

TYPE 15 **Matrix Operations and Simplification**

Given: $A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix}$, and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

To Find: The value of matrix expression $3A^2 - 2B + I$

Solution:

First evaluate $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2(2)+(-1)(3) & 2(-1)+(-1)(2) \\ 3(2)+2(3) & 3(-1)+2(2) \end{bmatrix} \\ &= \begin{bmatrix} 4-3 & -2-2 \\ 6+6 & -3+4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix} \end{aligned}$$

Calculate $3A^2$:

$$\begin{aligned} 3A^2 &= 3 \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & -12 \\ 36 & 3 \end{bmatrix} \end{aligned}$$

Calculate $2B$:

$$\begin{aligned} 2B &= 2 \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 8 \\ -2 & 14 \end{bmatrix} \end{aligned}$$

Now evaluate $3A^2 - 2B + I$:

$$\begin{aligned} 3A^2 - 2B + I &= \begin{bmatrix} 3 & -12 \\ 36 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 8 \\ -2 & 14 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3-0+1 & -12-8+0 \\ 36-(-2)+0 & 3-14+1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & -20 \\ 38 & -10 \end{bmatrix} \end{aligned}$$

Final Answer

$3A^2 - 2B + I = \begin{bmatrix} 4 & -20 \\ 38 & -10 \end{bmatrix}$

Question 18

If $A = \begin{bmatrix} 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$, verify that $A(B+C) = AB+AC$.

★ **Likely Exam Question**

TYPE 13 Matrix Multiplication Properties and Proofs

Given: $A = \begin{bmatrix} 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

To Prove: Verify the distributive property $A(B+C) = AB + AC$

Concept / Formula Used: $A(B+C) = AB + AC$

Solution:

First calculate LHS by evaluating $B+C$:

$$\begin{aligned} B + C &= \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix} + \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 5+(-1) & 3+2 & 4+1 \\ 8+0 & 7+1 & 6+2 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 5 & 5 \\ 8 & 8 & 8 \end{bmatrix} \end{aligned}$$

Now compute LHS $= A(B+C)$:

$$\begin{aligned} A(B+C) &= \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 5 & 5 \\ 8 & 8 & 8 \end{bmatrix} \\ &= \begin{bmatrix} 2(4)+1(8) & 2(5)+1(8) & 2(5)+1(8) \end{bmatrix} \\ &= \begin{bmatrix} 8+8 & 10+8 & 10+8 \end{bmatrix} \\ &= \begin{bmatrix} 16 & 18 & 18 \end{bmatrix} \end{aligned}$$

Next calculate RHS by evaluating AB and AC separately:

$$\begin{aligned} AB &= \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 2(5)+1(8) & 2(3)+1(7) & 2(4)+1(6) \end{bmatrix} \\ &= \begin{bmatrix} 10+8 & 6+7 & 8+6 \end{bmatrix} \\ &= \begin{bmatrix} 18 & 13 & 14 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} AC &= \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2(-1)+1(0) & 2(2)+1(1) & 2(1)+1(2) \end{bmatrix} \\ &= \begin{bmatrix} -2+0 & 4+1 & 2+2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 5 & 4 \end{bmatrix} \end{aligned}$$

Summing $AB + AC$:

$$\begin{aligned} AB + AC &= \begin{bmatrix} 18 & 13 & 14 \end{bmatrix} + \begin{bmatrix} -2 & 5 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 18+(-2) & 13+5 & 14+4 \end{bmatrix} \\ &= \begin{bmatrix} 16 & 18 & 18 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$\text{LHS} = \text{RHS} = \begin{bmatrix} 16 & 18 & 18 \end{bmatrix}$$

Hence Proved

LHS = RHS, hence $A(B+C) = AB + AC$ is verified.

Question 19

(i) Solve for x and y, given that $\begin{bmatrix} 2 & -3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

(ii) Solve for x and y, given that $\begin{bmatrix} x & y \\ 3y & x \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$.

TYPE 11 **Solving Matrix Equations using Multiplication**

Given: (i) $\begin{bmatrix} 2 & -3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$
(ii) $\begin{bmatrix} x & y \\ 3y & x \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

To Find: The values of x and y for each part

Concept / Formula Used: x y

Solution:

Part (i): Multiply the matrices on the LHS:

$$\begin{bmatrix} 2x - 3y \\ x + y \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Equating corresponding entries yields two linear equations:

$$2x - 3y = 1 \tag{3}$$
$$x + y = 3 \tag{4}$$

From equation (4), express x in terms of y:

$$x = 3 - y$$

Substitute $x = 3 - y$ into equation (3):

$$\begin{aligned} 2(3 - y) - 3y &= 1 \\ 6 - 2y - 3y &= 1 \\ 6 - 5y &= 1 \\ -5y &= 1 - 6 \\ -5y &= -5 \\ y &= 1 \end{aligned}$$

Substitute $y = 1$ back into $x = 3 - y$:

$$\begin{aligned} x &= 3 - 1 \\ x &= 2 \end{aligned}$$

Part (ii): Multiply the matrices on the LHS:

$$\begin{bmatrix} x(1) + y(2) \\ 3y(1) + x(2) \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x + 2y \\ 2x + 3y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Equating corresponding elements:

$$x + 2y = 3 \tag{5}$$
$$2x + 3y = 5 \tag{6}$$

From equation (5):

$$x = 3 - 2y$$

Substitute $x = 3 - 2y$ into equation (6):

$$\begin{aligned} 2(3 - 2y) + 3y &= 5 \\ 6 - 4y + 3y &= 5 \\ 6 - y &= 5 \\ -y &= 5 - 6 \\ -y &= -1 \\ y &= 1 \end{aligned}$$

Substitute $y = 1$ into $x = 3 - 2y$:

$$\begin{aligned} x &= 3 - 2(1) \\ x &= 1 \end{aligned}$$

Final Answer

(i) $x = 2$, $y = 1$

(ii) $x = 1$, $y = 1$

Question 20

(i) If $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$, show that $A^2 - 5A - 14I = O$.

(ii) If $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$, find the value of $A^2 + 2A + 7I$.

TYPE 17 **Evaluating Matrix Polynomials**

Given: (i) $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$

(ii) $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$

To Prove: (i) Show that $A^2 - 5A - 14I = O$
(ii) Compute the value of $A^2 + 2A + 7I$

Concept / Formula Used: 2×2

Solution:

Part (i): Calculate A^2:

$$\begin{aligned} A^2 &= \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3(3) + (-5)(-4) & 3(-5) + (-5)(2) \\ -4(3) + 2(-4) & -4(-5) + 2(2) \end{bmatrix} \\ &= \begin{bmatrix} 9 + 20 & -15 - 10 \\ -12 - 8 & 20 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix} \end{aligned}$$

Calculate $5A$:

$$\begin{aligned} 5A &= 5 \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix} \end{aligned}$$

Calculate $14I$:

$$\begin{aligned} 14I &= 14 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix} \end{aligned}$$

Now evaluate $A^2 - 5A - 14I$:

$$\begin{aligned} A^2 - 5A - 14I &= \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix} - \begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix} - \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix} \\ &= \begin{bmatrix} 29 - 15 - 14 & -25 - (-25) - 0 \\ -20 - (-20) - 0 & 24 - 10 - 14 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \end{aligned}$$

Part (ii): Calculate A^2:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(4) & 1(2) + 2(1) \\ 4(1) + 1(4) & 4(2) + 1(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 8 & 2 + 2 \\ 4 + 4 & 8 + 1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 4 \\ 8 & 9 \end{bmatrix} \end{aligned}$$

Calculate $2A$ and $7I$:

$$2A = 2\begin{bmatrix}1 & 2\\4 & 1\end{bmatrix} = \begin{bmatrix}2 & 4\\8 & 2\end{bmatrix}$$
$$7I = 7\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} = \begin{bmatrix}7 & 0\\0 & 7\end{bmatrix}$$

Now evaluate $A^2 + 2A + 7I$:

$$\begin{aligned}A^2 + 2A + 7I &= \begin{bmatrix}9 & 4\\8 & 9\end{bmatrix} + \begin{bmatrix}2 & 4\\8 & 2\end{bmatrix} + \begin{bmatrix}7 & 0\\0 & 7\end{bmatrix}\\ &= \begin{bmatrix}9+2+7 & 4+4+0\\8+8+0 & 9+2+7\end{bmatrix}\\ &= \begin{bmatrix}18 & 8\\16 & 18\end{bmatrix}\end{aligned}$$

Hence Proved

Part (i) shows $A^2 - 5A - 14I = O$. Part (ii) yields $A^2 + 2A + 7I = \begin{bmatrix}18 & 8\\16 & 18\end{bmatrix}$.

Question 21

If $A = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$ and $B = \begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix}$, show that $(aA + bB)(aA - bB) = (a^2 + b^2)A$.

TYPE 13 **Matrix Multiplication Properties and Proofs**

Given: $A = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$, $B = \begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix}$

To Prove: $(aA + bB)(aA - bB) = (a^2 + b^2)A$

Concept / Formula Used: $A = I \qquad AB = BA = B \qquad A^2 = A = I$

Solution:

Notice that $A = I$, the 2×2 identity matrix.
First compute matrices $aA + bB$ and $aA - bB$:

$$\begin{aligned}aA + bB &= a\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} + b\begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix}\\ &= \begin{bmatrix}a & 0\\0 & a\end{bmatrix} + \begin{bmatrix}0 & b\\-b & 0\end{bmatrix}\\ &= \begin{bmatrix}a & b\\-b & a\end{bmatrix}\end{aligned}$$

$$\begin{aligned} aA - bB &= a\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} - b\begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix} \\ &= \begin{bmatrix}a & 0\\0 & a\end{bmatrix} - \begin{bmatrix}0 & b\\-b & 0\end{bmatrix} \\ &= \begin{bmatrix}a & -b\\b & a\end{bmatrix}\end{aligned}$$

Multiply LHS $= (aA + bB)(aA - bB)$:

$$\begin{aligned} \text{LHS} &= \begin{bmatrix}a & b\\-b & a\end{bmatrix}\begin{bmatrix}a & -b\\b & a\end{bmatrix} \\ &= \begin{bmatrix}a(a)+b(b) & a(-b)+b(a)\\-b(a)+a(b) & -b(-b)+a(a)\end{bmatrix} \\ &= \begin{bmatrix}a^2+b^2 & -ab+ab\\-ab+ab & b^2+a^2\end{bmatrix} \\ &= \begin{bmatrix}a^2+b^2 & 0\\0 & a^2+b^2\end{bmatrix}\end{aligned}$$

Factor out $(a^2 + b^2)$:

$$\begin{aligned} \text{LHS} &= (a^2+b^2)\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} \\ &= (a^2+b^2)A = \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence $(aA + bB)(aA - bB) = (a^2 + b^2)A$ is proved.

Question 22

If $A = \begin{bmatrix}4 & 2\\-1 & 1\end{bmatrix}$, prove that $(A - 2I)(A - 3I) = O$.

TYPE 17 **Evaluating Matrix Polynomials**

Given: $A = \begin{bmatrix}4 & 2\\-1 & 1\end{bmatrix}$ and $I = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$

To Prove: $(A - 2I)(A - 3I) = O$

Solution:

Compute $A-2I$:

$$\begin{aligned} A-2I &= \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 4-2 & 2-0 \\ -1-0 & 1-2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \end{aligned}$$

Compute $A-3I$:

$$\begin{aligned} A-3I &= \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 4-3 & 2-0 \\ -1-0 & 1-3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \end{aligned}$$

Now calculate LHS $= (A-2I)(A-3I)$:

$$\begin{aligned} \text{LHS} &= \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 2(1)+2(-1) & 2(2)+2(-2) \\ -1(1)+(-1)(-1) & -1(2)+(-1)(-2) \end{bmatrix} \\ &= \begin{bmatrix} 2-2 & 4-4 \\ -1+1 & -2+2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence $(A-2I)(A-3I) = O$ is proved.

Question 23

If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$, then find k so that $A^2 = kA - 2I$.

TYPE 11 **Solving Matrix Equations using Multiplication**

Given: $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$ and $A^2 = kA - 2I$

To Find: The value of scalar k

Concept / Formula Used: $A^2 \quad kA - 2I$

Solution:

Calculate $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 3(3) + (-2)(4) & 3(-2) + (-2)(-2) \\ 4(3) + (-2)(4) & 4(-2) + (-2)(-2) \end{bmatrix} \\ &= \begin{bmatrix} 9 - 8 & -6 + 4 \\ 12 - 8 & -8 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} \end{aligned}$$

Now compute the RHS $= kA - 2I$:

$$\begin{aligned} kA - 2I &= k \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3k & -2k \\ 4k & -2k \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3k - 2 & -2k \\ 4k & -2k - 2 \end{bmatrix} \end{aligned}$$

Equating A^2 to $kA - 2I$:

$$\begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} = \begin{bmatrix} 3k - 2 & -2k \\ 4k & -2k - 2 \end{bmatrix}$$

Comparing entry at position $(1, 2)$:

$$\begin{aligned} -2k &= -2 \\ k &= 1 \end{aligned}$$

Check for consistency with other entries:

$$\begin{aligned} \text{Entry } (1,1) : \quad 3(1) - 2 = 1 \quad \text{(Matches)} \\ \text{Entry } (2,1) : \quad 4(1) = 4 \quad \text{(Matches)} \\ \text{Entry } (2,2) : \quad -2(1) - 2 = -4 \quad \text{(Matches)} \end{aligned}$$

The value $k = 1$ is consistent across all entries.

Final Answer

$k = 1$

Question 24 (i)

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, show that $f(A) = O$, where $f(x) = x^2 - 2x - 3$.

TYPE 17 **Evaluating Matrix Polynomials**

To Prove: $f(A) = A^2 - 2A - 3I = O$

Concept / Formula Used: $f(x) = a_n x^n + \cdots + a_1 x + a_0 \qquad f(A) = a_n A^n + \cdots + a_1 A + a_0 I \qquad I \qquad A$

Solution:

Given matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and identity matrix $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

First, evaluate A^2:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (1)(1) + (2)(2) & (1)(2) + (2)(1) \\ (2)(1) + (1)(2) & (2)(2) + (1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1+4 & 2+2 \\ 2+2 & 4+1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \end{aligned}$$

Next, evaluate $2A$:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix} \end{aligned}$$

Next, evaluate $3I$:

$$\begin{aligned} 3I &= 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \end{aligned}$$

Now, substitute A^2, $2A$, and $3I$ into $f(A) = A^2 - 2A - 3I$:

$$\begin{aligned} f(A) &= \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 5-2-3 & 4-4-0 \\ 4-4-0 & 5-2-3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ &= O \end{aligned}$$

Hence Proved

LHS = RHS, hence $f(A) = O$ is proved.

Question 24 (ii)

If $A = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$, find $f(A)$, where $f(x) = x^2 - 2x + 3$.

TYPE 17 **Evaluating Matrix Polynomials**

Given: $A = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$, $f(x) = x^2 - 2x + 3$

To Find: $f(A) = A^2 - 2A + 3I$

Concept / Formula Used: $f(A) = A^2 - 2A + 3I \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Solution:

First, compute A^2:

$$\begin{aligned} A^2 &= \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (-1)(-1) + (2)(3) & (-1)(2) + (2)(1) \\ (3)(-1) + (1)(3) & (3)(2) + (1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 6 & -2 + 2 \\ -3 + 3 & 6 + 1 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \end{aligned}$$

Next, compute $2A$:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 4 \\ 6 & 2 \end{bmatrix} \end{aligned}$$

Next, compute $3I$:

$$\begin{aligned} 3I &= 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \end{aligned}$$

Substitute these into $f(A) = A^2 - 2A + 3I$:

$$\begin{aligned} f(A) &= \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} - \begin{bmatrix} -2 & 4 \\ 6 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 7-(-2)+3 & 0-4+0 \\ 0-6+0 & 7-2+3 \end{bmatrix} \\ &= \begin{bmatrix} 7+2+3 & -4 \\ -6 & 5+3 \end{bmatrix} \\ &= \begin{bmatrix} 12 & -4 \\ -6 & 8 \end{bmatrix} \end{aligned}$$

Final Answer

$f(A) = \begin{bmatrix} 12 & -4 \\ -6 & 8 \end{bmatrix}$

Question 25

If $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then:

(i) find λ, μ so that $A^2 = \lambda A + \mu I$

(ii) prove that $A^3 - 4A^2 + A = O$

TYPE 17 **Evaluating Matrix Polynomials**

Given: $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

To Find: (i) λ, μ such that $A^2 = \lambda A + \mu I$; (ii) Prove $A^3 - 4A^2 + A = O$

Solution:

Part (i): First, compute A^2:

$$\begin{aligned} A^2 &= \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (2)(2)+(3)(1) & (2)(3)+(3)(2) \\ (1)(2)+(2)(1) & (1)(3)+(2)(2) \end{bmatrix} \\ &= \begin{bmatrix} 4+3 & 6+6 \\ 2+2 & 3+4 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} \end{aligned}$$

Now, express $\lambda A + \mu I$:

$$\begin{aligned}\lambda A + \mu I &= \lambda \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} + \mu \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2\lambda & 3\lambda \\ \lambda & 2\lambda \end{bmatrix} + \begin{bmatrix} \mu & 0 \\ 0 & \mu \end{bmatrix} \\ &= \begin{bmatrix} 2\lambda + \mu & 3\lambda \\ \lambda & 2\lambda + \mu \end{bmatrix}\end{aligned}$$

Equating $A^2 = \lambda A + \mu I$:

$$\begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 2\lambda + \mu & 3\lambda \\ \lambda & 2\lambda + \mu \end{bmatrix}$$

Equating corresponding elements:

$$\begin{aligned}\lambda &= 4 \\ 3\lambda &= 12 \implies \lambda = 4 \\ 2\lambda + \mu &= 7\end{aligned}$$

Substitute $\lambda = 4$ into $2\lambda + \mu = 7$:

$$\begin{aligned}2(4) + \mu &= 7 \\ 8 + \mu &= 7 \\ \mu &= 7 - 8 \\ \mu &= -1\end{aligned}$$

Part (ii): From Part (i), we have $A^2 = 4A - I$. Rearranging this equation gives:

$$A^2 - 4A + I = O$$

Multiplying both sides of the equation by A:

$$\begin{aligned}A(A^2 - 4A + I) &= A \cdot O \\ A^3 - 4A^2 + A \cdot I &= O \\ A^3 - 4A^2 + A &= O\end{aligned}$$

Final Answer

(i) $\lambda = 4, \quad \mu = -1$

(ii) $A^3 - 4A^2 + A = O$ is proved.

Question 26

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, verify that $A^2 - 4A - 5I = O$.

TYPE 17 Evaluating Matrix Polynomials

Given: $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

To Prove: $A^2 - 4A - 5I = O$

Concept / Formula Used: $(AB)_{ij} = \sum_k a_{ik}b_{kj}$

Solution:

First, calculate A^2:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} \end{aligned}$$

Next, calculate $4A$:

$$4A = 4\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix}$$

Next, calculate $5I$:

$$5I = 5\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Now compute $LHS = A^2 - 4A - 5I$:

$$\begin{aligned} LHS &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 9-4-5 & 8-8-0 & 8-8-0 \\ 8-8-0 & 9-4-5 & 8-8-0 \\ 8-8-0 & 8-8-0 & 9-4-5 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= O = RHS \end{aligned}$$

Hence Proved

LHS = RHS, hence $A^2 - 4A - 5I = O$ is verified.

Question 27

(i) Let A and B be square matrices of same order. Does $(A+B)^2 = A^2+2AB+B^2$ hold? If not, why?

(ii) Let A and B be square matrices of order 3×3. Is $(AB)^2 = A^2B^2$? Give reason.

TYPE 13 **Matrix Multiplication Properties and Proofs**

Given: A, B are square matrices of the same order.

To Find: Determine whether $(A+B)^2 = A^2+2AB+B^2$ and $(AB)^2 = A^2B^2$ hold, with reasons.

Concept / Formula Used: $AB \neq BA$

Solution:

Part (i): Expand the expression $(A+B)^2$:

$$\begin{aligned}(A+B)^2 &= (A+B)(A+B)\\ &= A(A+B)+B(A+B)\\ &= A^2+AB+BA+B^2\end{aligned}$$

For $(A+B)^2$ to equal $A^2+2AB+B^2$, we must have:

$$AB+BA = 2AB \implies BA = AB$$

Since matrix multiplication is non-commutative in general ($AB \neq BA$), $(A+B)^2 = A^2+2AB+B^2$ does NOT hold in general.

Part (ii): Expand the expressions $(AB)^2$ and A^2B^2:

$$\begin{aligned}(AB)^2 &= (AB)(AB) = A(BA)B\\ A^2B^2 &= (AA)(BB) = A(AB)B\end{aligned}$$

For $(AB)^2$ to equal A^2B^2, we must have:

$$A(BA)B = A(AB)B \impliedby BA = AB$$

Since matrix multiplication is not commutative in general, $BA \neq AB$, so $(AB)^2 = A^2B^2$ does NOT hold in general unless $AB = BA$.

Hence Proved

(i) No, because matrix multiplication is not commutative in general ($AB \neq BA$).

(ii) No, because $(AB)^2 = ABAB$ whereas $A^2B^2 = AABB$, which are equal only if $AB = BA$.

Question 28

If A and B are square matrices of the same order such that $AB = BA$, prove that:
(i) $(A+B)(A-B) = A^2 - B^2$
(ii) $(A+B)^2 = A^2 + 2AB + B^2$
(iii) $(A-B)^2 = A^2 - 2AB + B^2$
(iv) $(A+B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$
(v) $(A-B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$

TYPE 13 **Matrix Multiplication Properties and Proofs**

Given: A and B are square matrices with $AB = BA$.

To Prove: Prove identity subparts (i) through (v).

Concept / Formula Used: $A(B+C) = AB + AC \qquad AB = BA$

Solution:

Part (i):

$$\begin{aligned} LHS &= (A+B)(A-B) = A(A-B) + B(A-B) \\ &= A^2 - AB + BA - B^2 \end{aligned}$$

Since $AB = BA$, we have $-AB + BA = O$:

$$LHS = A^2 - AB + AB - B^2 = A^2 - B^2 = RHS$$

Part (ii):

$$\begin{aligned} LHS &= (A+B)^2 = (A+B)(A+B) = A(A+B) + B(A+B) \\ &= A^2 + AB + BA + B^2 \end{aligned}$$

Using $BA = AB$:

$$LHS = A^2 + AB + AB + B^2 = A^2 + 2AB + B^2 = RHS$$

Part (iii):

$$\begin{aligned} LHS &= (A-B)^2 = (A-B)(A-B) = A(A-B) - B(A-B) \\ &= A^2 - AB - BA + B^2 \end{aligned}$$

Using $BA = AB$:

$$LHS = A^2 - AB - AB + B^2 = A^2 - 2AB + B^2 = RHS$$

Part (iv): Using $(A+B)^2 = A^2 + 2AB + B^2$ from part (ii):

$$\begin{aligned} LHS &= (A+B)^3 = (A+B)^2(A+B) \\ &= (A^2 + 2AB + B^2)(A+B) \\ &= A^2(A+B) + 2AB(A+B) + B^2(A+B) \\ &= A^3 + A^2B + 2ABA + 2AB^2 + B^2A + B^3 \end{aligned}$$

Since $AB = BA$, we have $ABA = A(AB) = A(BA) = A^2B$ and $B^2A = B(BA) = B(AB) = (BA)B = (AB)B = AB^2$:

$$\begin{aligned} LHS &= A^3 + A^2B + 2A^2B + 2AB^2 + AB^2 + B^3 \\ &= A^3 + 3A^2B + 3AB^2 + B^3 = RHS \end{aligned}$$

Part (v): Using $(A - B)^2 = A^2 - 2AB + B^2$ from part (iii):

$$\begin{aligned} LHS &= (A - B)^3 = (A - B)^2(A - B) \\ &= (A^2 - 2AB + B^2)(A - B) \\ &= A^2(A - B) - 2AB(A - B) + B^2(A - B) \\ &= A^3 - A^2B - 2ABA + 2AB^2 + B^2A - B^3 \end{aligned}$$

Using $ABA = A^2B$ and $B^2A = AB^2$:

$$\begin{aligned} LHS &= A^3 - A^2B - 2A^2B + 2AB^2 + AB^2 - B^3 \\ &= A^3 - 3A^2B + 3AB^2 - B^3 = RHS \end{aligned}$$

Hence Proved

All five identities are proved using $AB = BA$.

Question 29

If $P = \begin{bmatrix} p & q \\ -q & p \end{bmatrix}$ and $Q = \begin{bmatrix} r & s \\ -s & r \end{bmatrix}$, prove that $PQ = QP$ for all values of p, q, r and s.

TYPE 13 **Matrix Multiplication Properties and Proofs**

Given: $P = \begin{bmatrix} p & q \\ -q & p \end{bmatrix}$, $Q = \begin{bmatrix} r & s \\ -s & r \end{bmatrix}$

To Prove: $PQ = QP$

Concept / Formula Used: $(AB)_{ij} = \sum_k a_{ik}b_{kj}$

Solution:

First, compute $LHS = PQ$:

$$\begin{aligned} PQ &= \begin{bmatrix} p & q \\ -q & p \end{bmatrix} \begin{bmatrix} r & s \\ -s & r \end{bmatrix} \\ &= \begin{bmatrix} (p)(r) + (q)(-s) & (p)(s) + (q)(r) \\ (-q)(r) + (p)(-s) & (-q)(s) + (p)(r) \end{bmatrix} \\ &= \begin{bmatrix} pr - qs & ps + qr \\ -qr - ps & -qs + pr \end{bmatrix} \\ &= \begin{bmatrix} pr - qs & ps + qr \\ -(ps + qr) & pr - qs \end{bmatrix} \end{aligned}$$

Next, compute $RHS = QP$:

$$\begin{aligned}
QP &= \begin{bmatrix} r & s \\ -s & r \end{bmatrix} \begin{bmatrix} p & q \\ -q & p \end{bmatrix} \\
&= \begin{bmatrix} (r)(p) + (s)(-q) & (r)(q) + (s)(p) \\ (-s)(p) + (r)(-q) & (-s)(q) + (r)(p) \end{bmatrix} \\
&= \begin{bmatrix} rp - sq & rq + sp \\ -sp - rq & -sq + rp \end{bmatrix} \\
&= \begin{bmatrix} pr - qs & ps + qr \\ -(ps + qr) & pr - qs \end{bmatrix}
\end{aligned}$$

Comparing LHS and RHS:

$$LHS = \begin{bmatrix} pr - qs & ps + qr \\ -(ps + qr) & pr - qs \end{bmatrix} = RHS$$

Hence Proved

LHS = RHS, hence $PQ = QP$ for all values of p, q, r and s.

Question 30

Find x, if:

(i) $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = O$

(ii) $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = O$

TYPE 11 Solving Matrix Equations using Multiplication

Given: Matrix equations involving x equal to zero matrix O.

To Find: The value(s) of x.

Concept / Formula Used: $(AB)C = A(BC)$

Solution:

Part (i): First, multiply the first two matrices (1×3 and 3×3):

$$\begin{aligned}
\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} &= \begin{bmatrix} 1(1) + x(2) + 1(15) & 1(3) + x(5) + 1(3) & 1(2) + x(1) + 1(2) \end{bmatrix} \\
&= \begin{bmatrix} 1 + 2x + 15 & 3 + 5x + 3 & 2 + x + 2 \end{bmatrix} \\
&= \begin{bmatrix} 2x + 16 & 5x + 6 & x + 4 \end{bmatrix}
\end{aligned}$$

Now, multiply this result by the third matrix $\begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix}$:

$$\begin{bmatrix} 2x+16 & 5x+6 & x+4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = O$$
$$\begin{bmatrix} (2x+16)(1) + (5x+6)(2) + (x+4)(x) \end{bmatrix} = [0]$$

Expanding the equation:

$$2x + 16 + 10x + 12 + x^2 + 4x = 0$$
$$x^2 + (2x + 10x + 4x) + (16 + 12) = 0$$
$$x^2 + 16x + 28 = 0$$

Factoring the quadratic equation:

$$x^2 + 14x + 2x + 28 = 0$$
$$x(x+14) + 2(x+14) = 0$$
$$(x+14)(x+2) = 0$$

Thus, $x = -14$ or $x = -2$.

Part (ii): First, multiply the first two matrices:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1(1)+2(2)+1(1) & 1(2)+2(0)+1(0) & 1(0)+2(1)+1(2) \end{bmatrix}$$
$$= \begin{bmatrix} 1+4+1 & 2+0+0 & 0+2+2 \end{bmatrix}$$
$$= \begin{bmatrix} 6 & 2 & 4 \end{bmatrix}$$

Now, multiply by the third matrix:

$$\begin{bmatrix} 6 & 2 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = O$$
$$\begin{bmatrix} 6(0) + 2(2) + 4(x) \end{bmatrix} = [0]$$
$$0 + 4 + 4x = 0$$
$$4x = -4$$
$$x = -1$$

Final Answer

(i) $x = -2$ or $x = -14$

(ii) $x = -1$

Question 31

If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ and $f(x) = x^2 - 4x + 7$, show that $f(A) = O$. Use this to find A^3 and A^5.

TYPE 18 **Finding Higher Powers of Matrices**

★ **Likely Exam Question**

Given: $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$, $f(x) = x^2 - 4x + 7$

To Find: Verify $f(A) = O$, and find A^3 and A^5.

Concept / Formula Used: $A^2 - 4A + 7I = O \qquad A^2 = 4A - 7I$

Solution:

First, calculate A^2:

$$\begin{aligned} A^2 &= \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (2)(2) + (3)(-1) & (2)(3) + (3)(2) \\ (-1)(2) + (2)(-1) & (-1)(3) + (2)(2) \end{bmatrix} \\ &= \begin{bmatrix} 4 - 3 & 6 + 6 \\ -2 - 2 & -3 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} \end{aligned}$$

Now calculate $f(A) = A^2 - 4A + 7I$:

$$\begin{aligned} f(A) &= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - 4 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 1 - 8 + 7 & 12 - 12 + 0 \\ -4 - (-4) + 0 & 1 - 8 + 7 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \end{aligned}$$

Thus, $f(A) = O$ is shown.

To find A^3, use $A^2 = 4A - 7I$:

$$\begin{aligned} A^3 &= A \cdot A^2 = A(4A - 7I) \\ &= 4A^2 - 7A \\ &= 4(4A - 7I) - 7A \\ &= 16A - 28I - 7A \\ &= 9A - 28I \end{aligned}$$

Substitute matrix A and I:

$$\begin{aligned} A^3 &= 9\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 28\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 18 & 27 \\ -9 & 18 \end{bmatrix} - \begin{bmatrix} 28 & 0 \\ 0 & 28 \end{bmatrix} \\ &= \begin{bmatrix} 18-28 & 27-0 \\ -9-0 & 18-28 \end{bmatrix} = \begin{bmatrix} -10 & 27 \\ -9 & -10 \end{bmatrix} \end{aligned}$$

To find A^5, first find A^4:

$$\begin{aligned} A^4 &= A \cdot A^3 = A(9A - 28I) = 9A^2 - 28A \\ &= 9(4A - 7I) - 28A = 36A - 63I - 28A \\ &= 8A - 63I \end{aligned}$$

Now, find A^5:

$$\begin{aligned} A^5 &= A \cdot A^4 = A(8A - 63I) = 8A^2 - 63A \\ &= 8(4A - 7I) - 63A = 32A - 56I - 63A \\ &= -31A - 56I \end{aligned}$$

Substitute matrix A and I:

$$\begin{aligned} A^5 &= -31\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 56\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -62 & -93 \\ 31 & -62 \end{bmatrix} - \begin{bmatrix} 56 & 0 \\ 0 & 56 \end{bmatrix} \\ &= \begin{bmatrix} -62-56 & -93-0 \\ 31-0 & -62-56 \end{bmatrix} = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix} \end{aligned}$$

Final Answer

$$A^3 = \begin{bmatrix} -10 & 27 \\ -9 & -10 \end{bmatrix}$$

$$A^5 = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}$$

Question 32

If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, prove that $A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$, for all positive integers n.

TYPE 19 Mathematical Induction on Matrices

★ **Likely Exam Question**

To Prove: $P(n) : A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$ for all $n \in \mathbb{N}$

Concept / Formula Used: $P(1) \qquad P(k) \qquad P(k+1)$

Solution:

Let the given statement be $P(n) : A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$.

Step 1: Base Case ($n = 1$)

$$LHS = A^1 = A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$
$$RHS = \begin{bmatrix} 2^{1-1} & 2^{1-1} \\ 2^{1-1} & 2^{1-1} \end{bmatrix} = \begin{bmatrix} 2^0 & 2^0 \\ 2^0 & 2^0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Thus, $P(1)$ is true.

Step 2: Inductive Hypothesis Assume $P(k)$ is true for some positive integer k, i.e.,

$$A^k = \begin{bmatrix} 2^{k-1} & 2^{k-1} \\ 2^{k-1} & 2^{k-1} \end{bmatrix}$$

Step 3: Inductive Step We need to show that $P(k+1)$ is true, i.e., $A^{k+1} = \begin{bmatrix} 2^k & 2^k \\ 2^k & 2^k \end{bmatrix}$.

$$\begin{aligned} A^{k+1} &= A^k \cdot A \\ &= \begin{bmatrix} 2^{k-1} & 2^{k-1} \\ 2^{k-1} & 2^{k-1} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2^{k-1}(1) + 2^{k-1}(1) & 2^{k-1}(1) + 2^{k-1}(1) \\ 2^{k-1}(1) + 2^{k-1}(1) & 2^{k-1}(1) + 2^{k-1}(1) \end{bmatrix} \\ &= \begin{bmatrix} 2 \cdot 2^{k-1} & 2 \cdot 2^{k-1} \\ 2 \cdot 2^{k-1} & 2 \cdot 2^{k-1} \end{bmatrix} \\ &= \begin{bmatrix} 2^k & 2^k \\ 2^k & 2^k \end{bmatrix} \end{aligned}$$

This shows that $P(k+1)$ is true whenever $P(k)$ is true. By the Principle of Mathematical Induction, $P(n)$ is true for all positive integers n.

Hence Proved

Result holds for all positive integers n.

Question 33

(i) Find a 2×2 matrix B such that $B \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = 6I_2$.

(ii) If $A = \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$, find matrix B such that $BA = I$.

TYPE 20 **Solving for Unknown Matrix**

Given: Matrix equations involving unknown matrix B.

To Find: Matrix B in both cases.

Solution:

Part (i): Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Given $B \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$
$$\begin{bmatrix} a+b & -2a+4b \\ c+d & -2c+4d \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

Equating elements for first row:

$$a + b = 6 \tag{7}$$
$$-2a + 4b = 0 \implies 2a = 4b \implies a = 2b \tag{8}$$

Substitute $a = 2b$ into equation (1):

$$2b + b = 6 \implies 3b = 6 \implies b = 2$$
$$a = 2(2) = 4$$

Equating elements for second row:

$$c + d = 0 \implies c = -d \tag{9}$$
$$-2c + 4d = 6 \tag{10}$$

Substitute $c = -d$ into equation (4):

$$-2(-d) + 4d = 6 \implies 6d = 6 \implies d = 1$$
$$c = -1$$

Thus, $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$.

Part (ii): Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Given $BA = I$:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$\begin{bmatrix} 2a+3b & a+2b \\ 2c+3d & c+2d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Equating corresponding elements:

$$2a + 3b = 1 \tag{11}$$
$$a + 2b = 0 \implies a = -2b \tag{12}$$

Substitute $a = -2b$ into (5):

$$2(-2b) + 3b = 1 \implies -4b + 3b = 1 \implies -b = 1 \implies b = -1$$
$$a = -2(-1) = 2$$

Also,

$$2c + 3d = 0 \tag{13}$$
$$c + 2d = 1 \implies c = 1 - 2d \tag{14}$$

Substitute $c = 1 - 2d$ into (7):

$$2(1-2d) + 3d = 0$$
$$2 - 4d + 3d = 0 \implies 2 - d = 0 \implies d = 2$$
$$c = 1 - 2(2) = 1 - 4 = -3$$

Thus, $B = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$.

Final Answer

(i) $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}, \quad (ii) B = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

Question 34

Find the matrix A such that $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} 3 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$.

TYPE 20 Solving for Unknown Matrix

Given: $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} 3 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$

To Find: The matrix A.

Concept / Formula Used: Compatibility for the product: since the left factor is 2×2 and the product is 2×3, A must be a 2×3 matrix.

Solution:

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$, since the product $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} A$ must be 2×3, so A is 2×3.

Substituting into the given equation:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} 3 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$$

Multiplying the matrices on the left-hand side:

$$\begin{bmatrix} a+d & b+e & c+f \\ d & e & f \end{bmatrix} = \begin{bmatrix} 3 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$$

Equating corresponding elements:

$$\begin{aligned} a+d &= 3 \quad \text{— (1)} \\ b+e &= 3 \quad \text{— (2)} \\ c+f &= 5 \quad \text{— (3)} \\ d &= 1 \quad \text{— (4)} \\ e &= 0 \quad \text{— (5)} \\ f &= 1 \quad \text{— (6)} \end{aligned}$$

From equation (4), $d = 1$; substituting into equation (1):

$$\begin{aligned} a+1 &= 3 \\ a &= 2 \end{aligned}$$

From equation (5), $e = 0$; substituting into equation (2):

$$\begin{aligned} b+0 &= 3 \\ b &= 3 \end{aligned}$$

From equation (6), $f = 1$; substituting into equation (3):

$$\begin{aligned} c+1 &= 5 \\ c &= 4 \end{aligned}$$

Thus, $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 1 \end{bmatrix}$.

Final Answer

$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 1 \end{bmatrix}$

Common Mistake: Assuming A is 2×2 by default is a common slip – always determine the order of the unknown matrix from the compatibility of the given product first.

Question 35

Three students Ram, Mohan and Ankit go to a shop to buy stationary. Ram purchases 2 dozen notebooks, 1 dozen pens and 4 pencils, Mohan purchases 1 dozen notebook, 6 pens and 8 pencils and Ankit purchases 6 notebooks, 4 pens and 6 pencils. A notebook costs Rs. 45, a pen costs Rs. 13.50 and a pencil costs Rs. 4.50. Using matrix multiplication, find the total amount of bill by all the three students.

TYPE 21 Matrix Multiplication Word Problems

Given: Ram: 2 dozen notebooks, 1 dozen pens, 4 pencils. Mohan: 1 dozen notebooks, 6 pens, 8 pencils. Ankit: 6 notebooks, 4 pens, 6 pencils. Notebook = Rs. 45, pen = Rs. 13.50, pencil = Rs. 4.50.

To Find: Total amount of bill by all three students, using matrix multiplication.

Concept / Formula Used: Represent the quantities purchased as a matrix P (order 3×3) and the prices as a column matrix C (order 3×1); the product PC gives each student's total bill.

Solution:

Converting dozens to individual units: 1 dozen = 12 items.

Ram purchases: $2 \times 12 = 24$ notebooks, $1 \times 12 = 12$ pens, 4 pencils.
Mohan purchases: $1 \times 12 = 12$ notebooks, 6 pens, 8 pencils.
Ankit purchases: 6 notebooks, 4 pens, 6 pencils.

Let the quantity matrix (rows = students, columns = notebooks, pens, pencils) be

$$P = \begin{bmatrix} 24 & 12 & 4 \\ 12 & 6 & 8 \\ 6 & 4 & 6 \end{bmatrix}$$

and the price matrix (column vector, order 3×1) be

$$C = \begin{bmatrix} 45 \\ 13.50 \\ 4.50 \end{bmatrix}$$

Since P is 3×3 and C is 3×1, the product PC is defined and gives a 3×1 matrix of total bills:

$$PC = \begin{bmatrix} 24 & 12 & 4 \\ 12 & 6 & 8 \\ 6 & 4 & 6 \end{bmatrix} \begin{bmatrix} 45 \\ 13.50 \\ 4.50 \end{bmatrix}$$

Computing each entry (row-by-column):

$$\begin{aligned}\text{Ram's bill} &= 24(45) + 12(13.50) + 4(4.50)\\ &= 1080 + 162 + 18\\ &= 1260\end{aligned}$$

$$\begin{aligned}\text{Mohan's bill} &= 12(45) + 6(13.50) + 8(4.50)\\ &= 540 + 81 + 36\\ &= 657\end{aligned}$$

$$\begin{aligned}\text{Ankit's bill} &= 6(45) + 4(13.50) + 6(4.50)\\ &= 270 + 54 + 27\\ &= 351\end{aligned}$$

So,

$$PC = \begin{bmatrix} 1260 \\ 657 \\ 351 \end{bmatrix}$$

Total amount of bill by all three students:

$$\begin{aligned}\text{Total} &= 1260 + 657 + 351\\ &= 2268\end{aligned}$$

Final Answer

Ram: Rs. 1260, Mohan: Rs. 657, Ankit: Rs. 351. Total bill by all three students = Rs. 2268

Common Mistake: Forgetting to convert "dozen" to 12 units before building the quantity matrix is the most common slip in this problem.

3.4 – Transpose, Symmetric and Skew-Symmetric Matrices

Question 1

If $A = \begin{bmatrix} 2 & -3 & 0 \\ -1 & 4 & 5 \end{bmatrix}$, then find $(3A)'$.

TYPE 22 **Matrix Transpose and Properties**

Given: Matrix $A = \begin{bmatrix} 2 & -3 & 0 \\ -1 & 4 & 5 \end{bmatrix}$.

Concept / Formula Used: k A $(kA)' = kA'$

Solution:

Given matrix A of order 2×3:

$$A = \begin{bmatrix} 2 & -3 & 0 \\ -1 & 4 & 5 \end{bmatrix}$$

First, multiply matrix A by the scalar 3:

$$\begin{aligned} 3A &= 3\begin{bmatrix} 2 & -3 & 0 \\ -1 & 4 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 3(2) & 3(-3) & 3(0) \\ 3(-1) & 3(4) & 3(5) \end{bmatrix} \\ &= \begin{bmatrix} 6 & -9 & 0 \\ -3 & 12 & 15 \end{bmatrix} \end{aligned}$$

Now, taking the transpose $(3A)'$ by interchanging rows and columns:

$$\begin{aligned} (3A)' &= \begin{bmatrix} 6 & -9 & 0 \\ -3 & 12 & 15 \end{bmatrix}' \\ &= \begin{bmatrix} 6 & -3 \\ -9 & 12 \\ 0 & 15 \end{bmatrix} \end{aligned}$$

Final Answer

$(3A)' = \begin{bmatrix} 6 & -3 \\ -9 & 12 \\ 0 & 15 \end{bmatrix}$

Question 2

If $A = \begin{bmatrix} 2 & -1 & 5 \\ 4 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & -3 \end{bmatrix}$, find $A' + B'$.

TYPE 22 **Matrix Transpose and Properties**

Given: Matrices $A = \begin{bmatrix} 2 & -1 & 5 \\ 4 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & -3 \end{bmatrix}$.

Concept / Formula Used: A' A

Solution:

First, find the transpose of matrix A:

$$\begin{aligned} A' &= \begin{bmatrix} 2 & -1 & 5 \\ 4 & 0 & 3 \end{bmatrix}' \\ &= \begin{bmatrix} 2 & 4 \\ -1 & 0 \\ 5 & 3 \end{bmatrix} \end{aligned}$$

Next, find the transpose of matrix B:

$$\begin{aligned} B' &= \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & -3 \end{bmatrix}' \\ &= \begin{bmatrix} -2 & -1 \\ 3 & 2 \\ 1 & -3 \end{bmatrix} \end{aligned}$$

Now, add the transposed matrices A' and B' element-wise:

$$\begin{aligned} A' + B' &= \begin{bmatrix} 2 & 4 \\ -1 & 0 \\ 5 & 3 \end{bmatrix} + \begin{bmatrix} -2 & -1 \\ 3 & 2 \\ 1 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 2+(-2) & 4+(-1) \\ -1+3 & 0+2 \\ 5+1 & 3+(-3) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 3 \\ 2 & 2 \\ 6 & 0 \end{bmatrix} \end{aligned}$$

Final Answer

$A' + B' = \begin{bmatrix} 0 & 3 \\ 2 & 2 \\ 6 & 0 \end{bmatrix}$

Question 3

If $A = \begin{bmatrix} \sin x & -\cos x \\ \cos x & \sin x \end{bmatrix}, 0 < x < \frac{\pi}{2}$ and $A + A' = I$ where I is unity matrix, find the value of x.

TYPE 22 Matrix Transpose and Properties

★ Likely Exam Question

Given: Matrix $A = \begin{bmatrix} \sin x & -\cos x \\ \cos x & \sin x \end{bmatrix}$ with $0 < x < \dfrac{\pi}{2}$ and $A + A' = I$.

Concept / Formula Used: $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\quad A + A' \quad I$

Solution:

Find the transpose of A:

$$A' = \begin{bmatrix} \sin x & \cos x \\ -\cos x & \sin x \end{bmatrix}$$

Compute $A + A'$:

$$\begin{aligned} A + A' &= \begin{bmatrix} \sin x & -\cos x \\ \cos x & \sin x \end{bmatrix} + \begin{bmatrix} \sin x & \cos x \\ -\cos x & \sin x \end{bmatrix} \\ &= \begin{bmatrix} \sin x + \sin x & -\cos x + \cos x \\ \cos x - \cos x & \sin x + \sin x \end{bmatrix} \\ &= \begin{bmatrix} 2\sin x & 0 \\ 0 & 2\sin x \end{bmatrix} \end{aligned}$$

Given $A + A' = I_2$:

$$\begin{bmatrix} 2\sin x & 0 \\ 0 & 2\sin x \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Equating corresponding elements:

$$\begin{aligned} 2\sin x &= 1 \\ \sin x &= \frac{1}{2} \end{aligned}$$

Since $0 < x < \frac{\pi}{2}$, we solve for x:

$$x = \frac{\pi}{6}$$

Final Answer

$x = \dfrac{\pi}{6}$

Common Mistake: Watch out for element positions: $\sin x$ is on the principal diagonal, so adding $A + A'$ gives $2\sin x$, not $2\cos x$.

Question 4

If $A^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, then find $A^T - B^T$.

TYPE 22 **Matrix Transpose and Properties**

Given: $A^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$.

Concept / Formula Used: $B_{m\times n} \quad B^T_{n\times m}$

Solution:

The matrix A^T is already given as:

$$A^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$$

Now, find the transpose of B:

$$B^T = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}^T$$
$$= \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$$

Subtract B^T from A^T element-wise:

$$A^T - B^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$$
$$= \begin{bmatrix} 3-(-1) & 4-1 \\ -1-2 & 2-2 \\ 0-1 & 1-3 \end{bmatrix}$$
$$= \begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{bmatrix}$$

Final Answer

$$A^T - B^T = \begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{bmatrix}$$

Question 5

For what value of k the matrix $\begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix}$ is a skew-symmetric?

TYPE 23 Symmetric and Skew-Symmetric Matrices

★ Likely Exam Question

Given: Matrix $M = \begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix}$ is skew-symmetric.

Concept / Formula Used: $M \qquad M^T = -M \qquad a_{ij} = -a_{ji}$

Solution:

Let $M = \begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix}$. Find the transpose M^T:

$$M^T = \begin{bmatrix} 0 & -6 \\ k & 0 \end{bmatrix}$$

Find $-M$:

$$\begin{aligned} -M &= -\begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -k \\ 6 & 0 \end{bmatrix} \end{aligned}$$

Since M is skew-symmetric, $M^T = -M$:

$$\begin{bmatrix} 0 & -6 \\ k & 0 \end{bmatrix} = \begin{bmatrix} 0 & -k \\ 6 & 0 \end{bmatrix}$$

Equating off-diagonal elements:

$$\begin{aligned} -k = -6 &\implies k = 6 \\ k = 6 \end{aligned}$$

Final Answer

$k = 6$

Question 6 (i)

Show that $(A + A')$ is a symmetric matrix, if $A = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$.

TYPE 23 Symmetric and Skew-Symmetric Matrices

★ **Likely Exam Question**

To Prove: $(A + A')$ is a symmetric matrix.

Concept / Formula Used: $X \qquad X' = X$

Solution:

Given matrix A:

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$$

Find the transpose A':

$$A' = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

Compute $X = A + A'$:

$$\begin{aligned} X = A + A' &= \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 2+2 & 4+3 \\ 3+4 & 5+5 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 7 \\ 7 & 10 \end{bmatrix} \end{aligned}$$

Now find the transpose of X:

$$\begin{aligned} X' = (A + A')' &= \begin{bmatrix} 4 & 7 \\ 7 & 10 \end{bmatrix}' \\ &= \begin{bmatrix} 4 & 7 \\ 7 & 10 \end{bmatrix} \\ &= X \end{aligned}$$

Since $(A + A')' = A + A'$, $(A + A')$ is a symmetric matrix.

Hence Proved

$(A + A')$ is a symmetric matrix.

Question 6 (ii)

If $A = \begin{bmatrix} 5 & -1 \\ -2 & 6 \end{bmatrix}$, determine whether $A - A^T$ is symmetric or skew-symmetric.

TYPE 23 **Symmetric and Skew-Symmetric Matrices**

Given: Matrix $A = \begin{bmatrix} 5 & -1 \\ -2 & 6 \end{bmatrix}$.

Concept / Formula Used: Y $\quad Y^T = Y \quad Y^T = -Y$

Solution:

Given matrix A:

$$A = \begin{bmatrix} 5 & -1 \\ -2 & 6 \end{bmatrix}$$

Find the transpose A^T:

$$A^T = \begin{bmatrix} 5 & -2 \\ -1 & 6 \end{bmatrix}$$

Compute $Y = A - A^T$:

$$\begin{aligned} Y = A - A^T &= \begin{bmatrix} 5 & -1 \\ -2 & 6 \end{bmatrix} - \begin{bmatrix} 5 & -2 \\ -1 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 5-5 & -1-(-2) \\ -2-(-1) & 6-6 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \end{aligned}$$

Take transpose of Y:

$$Y^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Notice that:

$$-Y = -\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Since $Y^T = -Y$, matrix $A - A^T$ is skew-symmetric.

Final Answer

$A - A^T$ is a skew-symmetric matrix.

Question 7 (i)

If $A = \begin{bmatrix} 5 & a \\ b & 0 \end{bmatrix}$ and A is symmetric matrix, show that $a = b$.

TYPE 23 **Symmetric and Skew-Symmetric Matrices**

★ Likely Exam Question

Given: Matrix $A = \begin{bmatrix} 5 & a \\ b & 0 \end{bmatrix}$ is symmetric.

To Prove: $a = b$

Concept / Formula Used: $A \qquad A' = A$

Solution:

Find the transpose A':

$$A' = \begin{bmatrix} 5 & b \\ a & 0 \end{bmatrix}$$

Since A is given to be a symmetric matrix, we have $A' = A$:

$$\begin{bmatrix} 5 & b \\ a & 0 \end{bmatrix} = \begin{bmatrix} 5 & a \\ b & 0 \end{bmatrix}$$

Equating the element at position $(1,2)$ (or $(2,1)$):

$$b = a$$
$$a = b$$

Hence Proved

$a = b$

Question 7 (ii)

If $A = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$ and A is symmetric, then write the relation between x and y.

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: Matrix $A = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$ is symmetric.

Concept / Formula Used: $A \quad A' = A \quad a_{12} = a_{21}$

Solution:

Given $A = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$. Find transpose A':

$$A' = \begin{bmatrix} 5 & y \\ x & 0 \end{bmatrix}$$

Since A is symmetric, $A' = A$:

$$\begin{bmatrix} 5 & y \\ x & 0 \end{bmatrix} = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$$

Equating corresponding elements:

$$x = y$$

Final Answer

$x = y$

Question 7 (iii)

If the matrix $\begin{bmatrix} 6 & -x^2 \\ 2x - 15 & 10 \end{bmatrix}$ is symmetric, find the value(s) of x.

TYPE 23 Symmetric and Skew-Symmetric Matrices

★ Likely Exam Question

Given: Matrix $M = \begin{bmatrix} 6 & -x^2 \\ 2x-15 & 10 \end{bmatrix}$ is symmetric.

Concept / Formula Used: $a_{12} = a_{21}$

Solution:

Equating elements at positions $(1,2)$ and $(2,1)$:

$$-x^2 = 2x - 15$$

Rearranging all terms to one side:

$$x^2 + 2x - 15 = 0$$

Factorize the quadratic expression:

$$\begin{aligned} x^2 + 5x - 3x - 15 &= 0 \\ x(x+5) - 3(x+5) &= 0 \\ (x+5)(x-3) &= 0 \end{aligned}$$

Solving for x:

$$\begin{aligned} x + 5 = 0 &\implies x = -5 \\ x - 3 = 0 &\implies x = 3 \end{aligned}$$

Final Answer

$x = 3$ or $x = -5$

Question 8

If A is a square matrix, prove that $A'A$ is symmetric.

TYPE 24 **Algebraic Proofs of Symmetric and Skew-Symmetric Matrices**

To Prove: $A'A$ is a symmetric matrix for any square matrix A.

Concept / Formula Used: $(AB)' = B'A' \qquad ((A')') = A$

Solution:

Let $X = A'A$. To prove X is symmetric, we must show that $X' = X$. Take the transpose of X:

$$X' = (A'A)'$$

Using the reversal rule of transpose $(AB)' = B'A'$ where first matrix is A' and second is A:

$$X' = A'(A')'$$

Using the identity $(A')' = A$:

$$X' = A'A$$

Since $X' = X$, the matrix $A'A$ is symmetric.

Hence Proved

$A'A$ is a symmetric matrix.

Question 9

If $A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$ is written as $A = P + Q$, where P is a symmetric matrix and Q is a skew-symmetric matrix, then write the matrix P.

TYPE 25 **Decomposing a Matrix into Symmetric and Skew-Symmetric Parts**

Given: $A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$, with $A = P + Q$, P symmetric, Q skew-symmetric.

Concept / Formula Used: $A \quad P + Q \quad P = \frac{1}{2}(A + A') \quad Q = \frac{1}{2}(A - A')$

Solution:

Given matrix A:

$$A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$$

Find A':

$$A' = \begin{bmatrix} 3 & 7 \\ 5 & 9 \end{bmatrix}$$

The symmetric matrix component P is given by:

$$\begin{aligned} P &= \frac{1}{2}(A + A') \\ &= \frac{1}{2}\left(\begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix} + \begin{bmatrix} 3 & 7 \\ 5 & 9 \end{bmatrix}\right) \\ &= \frac{1}{2}\begin{bmatrix} 3+3 & 5+7 \\ 7+5 & 9+9 \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} 6 & 12 \\ 12 & 18 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix} \end{aligned}$$

Final Answer

$P = \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix}$

Question 10

If A, B are symmetric matrices and $AB = BA$, then show that AB is symmetric.

TYPE 24 **Algebraic Proofs of Symmetric and Skew-Symmetric Matrices**

Given: $A' = A$, $B' = B$, and $AB = BA$.

To Prove: AB is symmetric, i.e., $(AB)' = AB$.

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Given that A and B are symmetric matrices:

$$A' = A$$
$$B' = B$$

Consider the transpose of the matrix product AB:

$$(AB)' = B'A' \quad \text{(using reversal law of transpose)}$$

Substitute $B' = B$ and $A' = A$:

$$(AB)' = BA$$

Given that $AB = BA$, substituting $BA = AB$:

$$(AB)' = AB$$

Since $(AB)' = AB$, the matrix AB is symmetric.

Hence Proved

AB is a symmetric matrix.

Question 11

If A, B and AB are all symmetric matrices, then show that $AB = BA$.

TYPE 24 **Algebraic Proofs of Symmetric and Skew-Symmetric Matrices**

Given: $A' = A$, $B' = B$, and $(AB)' = AB$.

To Prove: $AB = BA$

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Since A and B are symmetric matrices:

$$A' = A$$
$$B' = B$$

Since AB is also a symmetric matrix:

$$(AB)' = AB$$

By the reversal property of matrix transpose:

$$(AB)' = B'A'$$

Substitute $B' = B$ and $A' = A$:

$$(AB)' = BA$$

Equating both expressions for $(AB)'$:

$$AB = BA$$

Hence Proved

$AB = BA$

Question 12

If A, B are skew-symmetric matrices and $AB = BA$, then show that AB is symmetric.

TYPE 24 **Algebraic Proofs of Symmetric and Skew-Symmetric Matrices**

Given: $A' = -A$, $B' = -B$, and $AB = BA$.

To Prove: AB is symmetric, i.e., $(AB)' = AB$.

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Given A and B are skew-symmetric:

$$A' = -A$$
$$B' = -B$$

Take the transpose of AB:

$$(AB)' = B'A' \quad \text{(by reversal law of transpose)}$$

Substitute $B' = -B$ and $A' = -A$:

$$\begin{aligned}(AB)' &= (-B)(-A)\\ &= BA\end{aligned}$$

Since $AB = BA$, we have:

$$(AB)' = AB$$

Since $(AB)' = AB$, AB is symmetric.

Hence Proved

AB is a symmetric matrix.

Question 13

If A, B are skew-symmetric matrices of same order and AB is symmetric, then show that $AB = BA$.

TYPE 24 **Algebraic Proofs of Symmetric and Skew-Symmetric Matrices**

Given: $A' = -A$, $B' = -B$, and $(AB)' = AB$.

To Prove: $AB = BA$

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Given that A and B are skew-symmetric matrices:

$$A' = -A$$
$$B' = -B$$

Since AB is symmetric, we have:

$$(AB)' = AB$$

Using the reversal property of transpose on LHS:

$$(AB)' = B'A'$$

Substitute $B' = -B$ and $A' = -A$:

$$\begin{aligned}(AB)' &= (-B)(-A)\\ &= BA\end{aligned}$$

Equating the two results for $(AB)'$:

$$AB = BA$$

Hence Proved

$AB = BA$

Question 14

Let $A = \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, verify that (i) $(A')' = A$ (ii) $(3A)' = 3A'$ (iii) $(A-B)' = A' - B'$ (iv) $(AB)' = B'A'$

TYPE 22 **Matrix Transpose and Properties**

Given: $A = \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$.

Concept / Formula Used: $(A')' = A$ $(kA)' = kA'$ $(A-B)' = A' - B'$ $(AB)' = B'A'$

Solution:

Find basic transposes first:

$$A' = \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix}, \quad B' = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

(i) Verification of $(A')' = A$:

$$\text{LHS} = (A')' = \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix}' = \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} = A = \text{RHS}$$

(ii) Verification of $(3A)' = 3A'$:

$$3A = 3\begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 15 & -3 \\ 18 & 21 \end{bmatrix}$$
$$\text{LHS} = (3A)' = \begin{bmatrix} 15 & 18 \\ -3 & 21 \end{bmatrix}$$
$$\text{RHS} = 3A' = 3\begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 15 & 18 \\ -3 & 21 \end{bmatrix}$$
$$\text{LHS} = \text{RHS}$$

(iii) Verification of $(A-B)' = A' - B'$:

$$A - B = \begin{bmatrix} 5-2 & -1-1 \\ 6-3 & 7-4 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 3 & 3 \end{bmatrix}$$
$$\text{LHS} = (A-B)' = \begin{bmatrix} 3 & 3 \\ -2 & 3 \end{bmatrix}$$
$$\text{RHS} = A' - B' = \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 5-2 & 6-3 \\ -1-1 & 7-4 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ -2 & 3 \end{bmatrix}$$
$$\text{LHS} = \text{RHS}$$

(iv) Verification of $(AB)' = B'A'$:

$$AB = \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5(2)+(-1)(3) & 5(1)+(-1)(4) \\ 6(2)+7(3) & 6(1)+7(4) \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 33 & 34 \end{bmatrix}$$

$$\text{LHS} = (AB)' = \begin{bmatrix} 7 & 33 \\ 1 & 34 \end{bmatrix}$$

$$\text{RHS} = B'A' = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 2(5)+3(-1) & 2(6)+3(7) \\ 1(5)+4(-1) & 1(6)+4(7) \end{bmatrix} = \begin{bmatrix} 7 & 33 \\ 1 & 34 \end{bmatrix}$$

$$\text{LHS} = \text{RHS}$$

Hence Proved

All four identities are verified.

Question 15

If $A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, then verify that (i) $(A')' = A$ (ii) $(A+B)' = A' + B'$ (iii) $(kB)' = kB'$, where k is any real number.

TYPE 22 **Matrix Transpose and Properties**

Given: $A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$.

Concept / Formula Used: 2×3 3×2

Solution:

Transposes of A and B:

$$A' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix}, \quad B' = \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix}$$

(i) Verify $(A')' = A$:

$$(A')' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix}' = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix} = A$$

(ii) Verify $(A+B)' = A' + B'$**:**

$$A+B = \begin{bmatrix} 3+2 & \sqrt{3}-1 & 2+2 \\ 4+1 & 2+2 & 0+4 \end{bmatrix} = \begin{bmatrix} 5 & \sqrt{3}-1 & 4 \\ 5 & 4 & 4 \end{bmatrix}$$

$$\text{LHS} = (A+B)' = \begin{bmatrix} 5 & 5 \\ \sqrt{3}-1 & 4 \\ 4 & 4 \end{bmatrix}$$

$$\text{RHS} = A' + B' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 3+2 & 4+1 \\ \sqrt{3}-1 & 2+2 \\ 2+2 & 0+4 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ \sqrt{3}-1 & 4 \\ 4 & 4 \end{bmatrix}$$

$$\text{LHS} = \text{RHS}$$

(iii) Verify $(kB)' = kB'$**:**

$$kB = \begin{bmatrix} 2k & -k & 2k \\ k & 2k & 4k \end{bmatrix}$$

$$\text{LHS} = (kB)' = \begin{bmatrix} 2k & k \\ -k & 2k \\ 2k & 4k \end{bmatrix}$$

$$\text{RHS} = kB' = k\begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2k & k \\ -k & 2k \\ 2k & 4k \end{bmatrix}$$

$$\text{LHS} = \text{RHS}$$

Hence Proved

All three properties are verified.

Question 16

If $A' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$, then find $(A+2B)'$.

TYPE 22 **Matrix Transpose and Properties**

Given: $A' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$.

Concept / Formula Used: $(A+2B)' = A' + 2B'$

Solution:

Method 1: Find matrix A first.

$$A = (A')' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}' = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}$$

Calculate $2B$:

$$2B = 2\begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 2 & 4 \end{bmatrix}$$

Compute $A + 2B$:

$$\begin{aligned} A + 2B &= \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} -2 & 0 \\ 2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} -2 + (-2) & 1 + 0 \\ 3 + 2 & 2 + 4 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 1 \\ 5 & 6 \end{bmatrix} \end{aligned}$$

Take transpose of $A + 2B$:

$$\begin{aligned} (A + 2B)' &= \begin{bmatrix} -4 & 1 \\ 5 & 6 \end{bmatrix}' \\ &= \begin{bmatrix} -4 & 5 \\ 1 & 6 \end{bmatrix} \end{aligned}$$

Final Answer

$(A + 2B)' = \begin{bmatrix} -4 & 5 \\ 1 & 6 \end{bmatrix}$

Question 17

If $A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$, verify that $(AB)' = B'A'$.

TYPE 22 **Matrix Transpose and Properties**

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$ (3×1) and $B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$ (1×3).

Concept / Formula Used: $(AB)' = B'A'$

Solution:

First calculate the product matrix AB (3×3):

$$\begin{aligned} AB &= \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1(-1) & 1(2) & 1(1) \\ -4(-1) & -4(2) & -4(1) \\ 3(-1) & 3(2) & 3(1) \end{bmatrix} \\ &= \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix} \end{aligned}$$

Now, find the LHS, $(AB)'$:

$$\text{LHS} = (AB)' = \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix}'$$
$$= \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$$

Next, find B' and A':

$$B' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad (3 \times 1)$$
$$A' = \begin{bmatrix} 1 & -4 & 3 \end{bmatrix} \quad (1 \times 3)$$

Calculate the RHS, $B'A'$:

$$\text{RHS} = B'A' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -4 & 3 \end{bmatrix}$$
$$= \begin{bmatrix} -1(1) & -1(-4) & -1(3) \\ 2(1) & 2(-4) & 2(3) \\ 1(1) & 1(-4) & 1(3) \end{bmatrix}$$
$$= \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$$

Comparing LHS and RHS:

$$\text{LHS} = \text{RHS}$$

Hence Proved

$(AB)' = B'A'$ is verified.

Common Mistake: Keep track of matrix dimensions: A is 3×1 and B is 1×3, making AB a 3×3 matrix.

Question 18

If $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 6 & 4 \\ 7 & 3 \end{bmatrix}$, then verify that (i) $(2A + B)' = 2A' + B'$ (ii) $(A - B)' = A' - B'$

TYPE 22 **Matrix Transpose and Properties**

Given: $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 6 & 4 \\ 7 & 3 \end{bmatrix}$.

Solution:

Find transposes A' and B' (2×3):

$$A' = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 1 & 6 \end{bmatrix}, \quad B' = \begin{bmatrix} 1 & 6 & 7 \\ 2 & 4 & 3 \end{bmatrix}$$

(i) Verification of $(2A + B)' = 2A' + B'$:

$$2A = 2\begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 8 & 2 \\ 10 & 12 \end{bmatrix}$$

$$2A + B = \begin{bmatrix} 2+1 & 4+2 \\ 8+6 & 2+4 \\ 10+7 & 12+3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 14 & 6 \\ 17 & 15 \end{bmatrix}$$

$$\text{LHS} = (2A + B)' = \begin{bmatrix} 3 & 14 & 17 \\ 6 & 6 & 15 \end{bmatrix}$$

$$2A' = 2\begin{bmatrix} 1 & 4 & 5 \\ 2 & 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 8 & 10 \\ 4 & 2 & 12 \end{bmatrix}$$

$$\text{RHS} = 2A' + B' = \begin{bmatrix} 2+1 & 8+6 & 10+7 \\ 4+2 & 2+4 & 12+3 \end{bmatrix} = \begin{bmatrix} 3 & 14 & 17 \\ 6 & 6 & 15 \end{bmatrix}$$

$$\text{LHS} = \text{RHS}$$

(ii) Verification of $(A - B)' = A' - B'$:

$$A - B = \begin{bmatrix} 1-1 & 2-2 \\ 4-6 & 1-4 \\ 5-7 & 6-3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -2 & -3 \\ -2 & 3 \end{bmatrix}$$

$$\text{LHS} = (A - B)' = \begin{bmatrix} 0 & -2 & -2 \\ 0 & -3 & 3 \end{bmatrix}$$

$$\text{RHS} = A' - B' = \begin{bmatrix} 1-1 & 4-6 & 5-7 \\ 2-2 & 1-4 & 6-3 \end{bmatrix} = \begin{bmatrix} 0 & -2 & -2 \\ 0 & -3 & 3 \end{bmatrix}$$

$$\text{LHS} = \text{RHS}$$

Hence Proved

Both identities are verified.

Question 19

If $A = \begin{bmatrix} 2 & 4 & 0 \\ 3 & 9 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 1 & 3 \end{bmatrix}$, verify that $(AB)' = B'A'$.

TYPE 22 Matrix Transpose and Properties

Given: $A = \begin{bmatrix} 2 & 4 & 0 \\ 3 & 9 & 6 \end{bmatrix}$ (2×3) and $B = \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 1 & 3 \end{bmatrix}$ (3×2).

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Compute product AB (2×2):

$$AB = \begin{bmatrix} 2 & 4 & 0 \\ 3 & 9 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 1 & 3 \end{bmatrix}$$
$$= \begin{bmatrix} 2(1)+4(2)+0(1) & 2(4)+4(8)+0(3) \\ 3(1)+9(2)+6(1) & 3(4)+9(8)+6(3) \end{bmatrix}$$
$$= \begin{bmatrix} 2+8+0 & 8+32+0 \\ 3+18+6 & 12+72+18 \end{bmatrix}$$
$$= \begin{bmatrix} 10 & 40 \\ 27 & 102 \end{bmatrix}$$

Find LHS $= (AB)'$:

$$\text{LHS} = (AB)' = \begin{bmatrix} 10 & 27 \\ 40 & 102 \end{bmatrix}$$

Find transposes B' (2×3) and A' (3×2):

$$B' = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 8 & 3 \end{bmatrix}, \quad A' = \begin{bmatrix} 2 & 3 \\ 4 & 9 \\ 0 & 6 \end{bmatrix}$$

Compute RHS $= B'A'$ (2×2):

$$\text{RHS} = B'A' = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 8 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 9 \\ 0 & 6 \end{bmatrix}$$
$$= \begin{bmatrix} 1(2)+2(4)+1(0) & 1(3)+2(9)+1(6) \\ 4(2)+8(4)+3(0) & 4(3)+8(9)+3(6) \end{bmatrix}$$
$$= \begin{bmatrix} 2+8+0 & 3+18+6 \\ 8+32+0 & 12+72+18 \end{bmatrix}$$
$$= \begin{bmatrix} 10 & 27 \\ 40 & 102 \end{bmatrix}$$

Comparing LHS and RHS:

$$\text{LHS} = \text{RHS}$$

Hence Proved

$(AB)' = B'A'$ is verified.

Question 20

If $A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 2 & 5 \\ 3 & 4 \end{bmatrix}$, find $(BA)'$.

TYPE 22 **Matrix Transpose and Properties**

Given: $A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ (2×2) and $B = \begin{bmatrix} -1 & 0 \\ 2 & 5 \\ 3 & 4 \end{bmatrix}$ (3×2).

Concept / Formula Used: $(BA)'$ BA $(BA)' = A'B'$

Solution:

Matrix B has order 3×2 and matrix A has order 2×2. The product BA is defined and will have order 3×2:

$$\begin{aligned} BA &= \begin{bmatrix} -1 & 0 \\ 2 & 5 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -1(3) + 0(-1) & -1(2) + 0(1) \\ 2(3) + 5(-1) & 2(2) + 5(1) \\ 3(3) + 4(-1) & 3(2) + 4(1) \end{bmatrix} \\ &= \begin{bmatrix} -3 + 0 & -2 + 0 \\ 6 - 5 & 4 + 5 \\ 9 - 4 & 6 + 4 \end{bmatrix} \\ &= \begin{bmatrix} -3 & -2 \\ 1 & 9 \\ 5 & 10 \end{bmatrix} \end{aligned}$$

Now take transpose of BA:

$$\begin{aligned} (BA)' &= \begin{bmatrix} -3 & -2 \\ 1 & 9 \\ 5 & 10 \end{bmatrix}' \\ &= \begin{bmatrix} -3 & 1 & 5 \\ -2 & 9 & 10 \end{bmatrix} \end{aligned}$$

Final Answer

$(BA)' = \begin{bmatrix} -3 & 1 & 5 \\ -2 & 9 & 10 \end{bmatrix}$

Question 21

If $A = \begin{bmatrix} \sin\alpha & \cos\alpha \\ -\cos\alpha & \sin\alpha \end{bmatrix}$, then verify that $A'A = I$.

TYPE 26 **Orthogonal Matrix Properties**

Given: $A = \begin{bmatrix} \sin\alpha & \cos\alpha \\ -\cos\alpha & \sin\alpha \end{bmatrix}$

To Prove: $A'A = I$

Concept / Formula Used: $(A')_{ij} = A_{ji} \qquad \sin^2\alpha + \cos^2\alpha = 1$

Solution:

By interchanging the rows and columns of matrix A, we obtain its transpose A':

$$A' = \begin{bmatrix} \sin\alpha & -\cos\alpha \\ \cos\alpha & \sin\alpha \end{bmatrix}$$

Now, computing the product $A'A$:

$$\begin{aligned} A'A &= \begin{bmatrix} \sin\alpha & -\cos\alpha \\ \cos\alpha & \sin\alpha \end{bmatrix} \begin{bmatrix} \sin\alpha & \cos\alpha \\ -\cos\alpha & \sin\alpha \end{bmatrix} \\ &= \begin{bmatrix} (\sin\alpha)(\sin\alpha) + (-\cos\alpha)(-\cos\alpha) & (\sin\alpha)(\cos\alpha) + (-\cos\alpha)(\sin\alpha) \\ (\cos\alpha)(\sin\alpha) + (\sin\alpha)(-\cos\alpha) & (\cos\alpha)(\cos\alpha) + (\sin\alpha)(\sin\alpha) \end{bmatrix} \\ &= \begin{bmatrix} \sin^2\alpha + \cos^2\alpha & \sin\alpha\cos\alpha - \cos\alpha\sin\alpha \\ \cos\alpha\sin\alpha - \sin\alpha\cos\alpha & \cos^2\alpha + \sin^2\alpha \end{bmatrix} \end{aligned}$$

Using the standard trigonometric identity $\sin^2\alpha + \cos^2\alpha = 1$:

$$\begin{aligned} A'A &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

Hence Proved

$A'A = I$, hence verified.

Question 22

If $A = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$, show that $AA' = I$.

TYPE 26 **Orthogonal Matrix Properties**

Given: $A = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$

To Prove: $AA' = I$

Concept / Formula Used: $(kA)' = kA' \qquad (AB)_{ij} = \sum_k a_{ik}b_{kj}$

Solution:

Taking the transpose of matrix A:

$$A' = \frac{1}{3}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{bmatrix}$$

Now, computing the product AA':

$$\begin{aligned}
AA' &= \left(\frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}\right)\left(\frac{1}{3}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{bmatrix}\right) \\
&= \frac{1}{9}\begin{bmatrix} 1(1)+(-2)(-2)+2(2) & 1(-2)+(-2)(1)+2(2) & 1(-2)+(-2)(-2)+2(-1) \\ -2(1)+1(-2)+2(2) & -2(-2)+1(1)+2(2) & -2(-2)+1(-2)+2(-1) \\ -2(1)+(-2)(-2)+(-1)(2) & -2(-2)+(-2)(1)+(-1)(2) & -2(-2)+(-2)(-2)+(-1)(-1) \end{bmatrix} \\
&= \frac{1}{9}\begin{bmatrix} 1+4+4 & -2-2+4 & -2+4-2 \\ -2-2+4 & 4+1+4 & 4-2-2 \\ -2+4-2 & 4-2-2 & 4+4+1 \end{bmatrix} \\
&= \frac{1}{9}\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
&= I
\end{aligned}$$

Hence Proved

$AA' = I$, hence proved.

Question 23

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & x \\ -2 & 2 & -1 \end{bmatrix}$ is a matrix satisfying $AA' = 9I_3$, find x.

TYPE 26 **Orthogonal Matrix Properties**

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & x \\ -2 & 2 & -1 \end{bmatrix}$ and $AA' = 9I_3$

To Find: Value of x

Concept / Formula Used: $AA' = 9I_3 \qquad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Solution:

The transpose of matrix A is given by:

$$A' = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & x & -1 \end{bmatrix}$$

Evaluating the matrix product AA':

$$\begin{aligned}
AA' &= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & x \\ -2 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & x & -1 \end{bmatrix} \\
&= \begin{bmatrix} 1(1)+2(2)+2(2) & 1(2)+2(1)+2(x) & 1(-2)+2(2)+2(-1) \\ 2(1)+1(2)+x(2) & 2(2)+1(1)+x(x) & 2(-2)+1(2)+x(-1) \\ -2(1)+2(2)+(-1)(2) & -2(2)+2(1)+(-1)(x) & -2(-2)+2(2)+(-1)(-1) \end{bmatrix} \\
&= \begin{bmatrix} 1+4+4 & 2+2+2x & -2+4-2 \\ 2+2+2x & 4+1+x^2 & -4+2-x \\ -2+4-2 & -4+2-x & 4+4+1 \end{bmatrix} \\
&= \begin{bmatrix} 9 & 4+2x & 0 \\ 4+2x & 5+x^2 & -2-x \\ 0 & -2-x & 9 \end{bmatrix}
\end{aligned}$$

We are given $AA' = 9I_3$:

$$\begin{bmatrix} 9 & 4+2x & 0 \\ 4+2x & 5+x^2 & -2-x \\ 0 & -2-x & 9 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Equating corresponding elements: From element $(1,2)$:

$$\begin{aligned} 4+2x &= 0 \\ 2x &= -4 \\ x &= -2 \end{aligned}$$

Verifying from element $(2,3)$:

$$\begin{aligned} -2-x &= 0 \\ x &= -2 \end{aligned}$$

Verifying from element $(2,2)$:

$$\begin{aligned} 5+x^2 &= 9 \\ 5+(-2)^2 &= 5+4 = 9 \quad \text{(Satisfied)} \end{aligned}$$

Final Answer

$x = -2$

Question 24

Find the integral value of x if $\begin{bmatrix} x & 4 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x & 4 & -1 \end{bmatrix}' = O$.

TYPE 22 **Matrix Transpose and Properties**

Given: $\begin{bmatrix} x & 4 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x & 4 & -1 \end{bmatrix}' = O$

To Find: Integral value of x

Concept / Formula Used: $(XA)X' = O \qquad ax^2 + bx + c = 0$

Solution:

The transpose of the row vector $\begin{bmatrix} x & 4 & -1 \end{bmatrix}$ is the column vector:

$$\begin{bmatrix} x & 4 & -1 \end{bmatrix}' = \begin{bmatrix} x \\ 4 \\ -1 \end{bmatrix}$$

First, compute the product of the first two matrices:

$$\begin{aligned}
\begin{bmatrix} x & 4 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & 2 & 4 \end{bmatrix} &= \begin{bmatrix} 2(x) + 1(4) + 2(-1) & 1(x) + 0(4) + 2(-1) & -1(x) + 0(4) + 4(-1) \end{bmatrix} \\
&= \begin{bmatrix} 2x + 4 - 2 & x + 0 - 2 & -x + 0 - 4 \end{bmatrix} \\
&= \begin{bmatrix} 2x + 2 & x - 2 & -x - 4 \end{bmatrix}
\end{aligned}$$

Now, multiply this resulting row vector by the column vector:

$$\begin{aligned}
\begin{bmatrix} 2x + 2 & x - 2 & -x - 4 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ -1 \end{bmatrix} &= [0] \\
(2x + 2)(x) + (x - 2)(4) + (-x - 4)(-1) &= 0 \\
2x^2 + 2x + 4x - 8 + x + 4 &= 0 \\
2x^2 + 7x - 4 &= 0
\end{aligned}$$

Factoring the quadratic equation:

$$\begin{aligned}
2x^2 + 8x - x - 4 &= 0 \\
2x(x + 4) - 1(x + 4) &= 0 \\
(2x - 1)(x + 4) &= 0
\end{aligned}$$

Solving for x:

$$2x - 1 = 0 \implies x = \frac{1}{2}$$
$$x + 4 = 0 \implies x = -4$$

Since x must be an integral value (an integer), we reject $x = \frac{1}{2}$.

Final Answer

$x = -4$

Common Mistake: Taking $x = \frac{1}{2}$ as part of the final answer without noticing the question explicitly asks for the integral value of x.

Question 25

Show that the matrix A, where $A = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, is a symmetric matrix.

TYPE 23 **Symmetric and Skew-Symmetric Matrices**

Given: $A = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$

To Prove: A is a symmetric matrix

Concept / Formula Used: $A \qquad A' = A$

Solution:

Find the transpose A' by interchanging the rows and columns of matrix A:

$$A' = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$$

Comparing matrix A' with matrix A:

$$A' = A$$

Hence Proved

Since $A' = A$, the matrix A is a symmetric matrix.

Question 26

If the matrix $\begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$ is symmetric, find the values of a and b.

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: $A = \begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$ is symmetric

To Find: Values of a and b

Concept / Formula Used: $a_{ij} = a_{ji} \quad i \quad j$

Solution:

Let $A = \begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$. Since A is a symmetric matrix, we have $A' = A$, which implies $a_{ij} = a_{ji}$ for all i, j.

Equating a_{12} and a_{21}:

$$2b = 3$$
$$b = \frac{3}{2}$$

Equating a_{13} and a_{31}:

$$3a = -2$$
$$a = -\frac{2}{3}$$

Checking a_{23} and a_{32}:

$$3 = 3 \quad \text{(which holds naturally)}$$

Final Answer

$a = -\frac{2}{3}, \quad b = \frac{3}{2}$

Question 27

If A and B are symmetric matrices of the same order, prove that $AB + BA$ is symmetric.

TYPE 24 Algebraic Proofs of Symmetric and Skew-Symmetric Matrices

Given: A and B are symmetric matrices of the same order, so $A' = A$ and $B' = B$

To Prove: $AB + BA$ is a symmetric matrix

Concept / Formula Used: $(X + Y)' = X' + Y' \qquad (XY)' = Y'X'$

Solution:

To prove that $AB + BA$ is symmetric, we must show that $(AB + BA)' = AB + BA$.

Taking the transpose of $(AB + BA)$:

$$(AB + BA)' = (AB)' + (BA)' \quad [\text{using } (X + Y)' = X' + Y']$$

Applying the reversal law of transposes $(XY)' = Y'X'$:

$$= B'A' + A'B'$$

Since A and B are symmetric matrices, substitute $A' = A$ and $B' = B$:

$$= BA + AB$$

Using commutativity of matrix addition:

$$= AB + BA$$

Thus, $(AB + BA)' = AB + BA$.

Hence Proved

Hence, $AB + BA$ is a symmetric matrix.

Question 28

If A and B are symmetric matrices such that AB and BA are both defined, then prove that $AB - BA$ is a skew-symmetric matrix.

TYPE 24 **Algebraic Proofs of Symmetric and Skew-Symmetric Matrices**

★ Likely Exam Question

Given: A and B are symmetric matrices of the same order, so $A' = A$ and $B' = B$

To Prove: $AB - BA$ is a skew-symmetric matrix

Concept / Formula Used: $X \qquad X' = -X \qquad (X - Y)' = X' - Y' \qquad (XY)' = Y'X'$

Solution:

To prove that $AB - BA$ is skew-symmetric, we must show that $(AB - BA)' = -(AB - BA)$.

Taking the transpose of $(AB - BA)$:

$$(AB - BA)' = (AB)' - (BA)' \quad [\text{using } (X - Y)' = X' - Y']$$

Applying the reversal law of transposes $(XY)' = Y'X'$:

$$= B'A' - A'B'$$

Since A and B are symmetric matrices, substitute $A' = A$ and $B' = B$:

$$= BA - AB$$

Factoring out a negative sign:

$$= -(AB - BA)$$

Thus, $(AB - BA)' = -(AB - BA)$.

Hence Proved

Hence, $AB - BA$ is a skew-symmetric matrix.

Question 29

Express each of the following matrices as the sum of a symmetric and a skew-symmetric matrices:

(i) $\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$

(ii) $\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$

TYPE 25 **Decomposing a Matrix into Symmetric and Skew-Symmetric Parts**

Given: Matrices $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ for part (i) and $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ for part (ii)

To Find: Decomposition $A = P + Q$ where $P' = P$ and $Q' = -Q$

Concept / Formula Used: $A \qquad A = P + Q \qquad P = \frac{1}{2}(A + A') \qquad Q = \frac{1}{2}(A - A')$

Solution:

Part (i): Let $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$.

1. Find the transpose A':

$$A' = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}$$

2. Compute the symmetric part $P = \frac{1}{2}(A + A')$:

$$A + A' = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 6 \\ 6 & -2 \end{bmatrix}$$

$$P = \frac{1}{2}\begin{bmatrix} 6 & 6 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix}$$

3. Compute the skew-symmetric part $Q = \frac{1}{2}(A - A')$:

$$A - A' = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}$$

$$Q = \frac{1}{2} \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

4. Writing $A = P + Q$:

$$\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

Part (ii): Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$.

1. Find the transpose A':

$$A' = \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix}$$

2. Compute the symmetric part $P = \frac{1}{2}(A + A')$:

$$A + A' = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & -2 \end{bmatrix}$$

$$P = \frac{1}{2} \begin{bmatrix} 6 & -3 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & -1 \end{bmatrix}$$

3. Compute the skew-symmetric part $Q = \frac{1}{2}(A - A')$:

$$A - A' = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix}$$

$$Q = \frac{1}{2} \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$$

4. Writing $A = P + Q$:

$$\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & -1 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$$

Final Answer

(i) $\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

(ii) $\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & -1 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$

Question 30

Express the following matrices as the sum of symmetric and skew-symmetric matrices:

(i) $\begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix}$

(ii) $\begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}$

TYPE 25 **Decomposing a Matrix into Symmetric and Skew-Symmetric Parts**

Given: Matrices $A = \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix}$ for part (i) and $A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}$ for part (ii)

To Find: Decomposition $A = P + Q$ where $P' = P$ and $Q' = -Q$

Concept / Formula Used: $A = P + Q \qquad P = \frac{1}{2}(A + A') \qquad Q = \frac{1}{2}(A - A')$

Solution:

Part (i): Let $A = \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix}$.

1. Find the transpose A':

$$A' = \begin{bmatrix} 3 & 4 & 0 \\ 2 & 1 & 6 \\ 5 & 3 & 7 \end{bmatrix}$$

2. Compute the symmetric matrix $P = \frac{1}{2}(A + A')$:

$$A + A' = \begin{bmatrix} 3+3 & 2+4 & 5+0 \\ 4+2 & 1+1 & 3+6 \\ 0+5 & 6+3 & 7+7 \end{bmatrix} = \begin{bmatrix} 6 & 6 & 5 \\ 6 & 2 & 9 \\ 5 & 9 & 14 \end{bmatrix}$$

$$P = \frac{1}{2}\begin{bmatrix} 6 & 6 & 5 \\ 6 & 2 & 9 \\ 5 & 9 & 14 \end{bmatrix} = \begin{bmatrix} 3 & 3 & \frac{5}{2} \\ 3 & 1 & \frac{9}{2} \\ \frac{5}{2} & \frac{9}{2} & 7 \end{bmatrix}$$

3. Compute the skew-symmetric matrix $Q = \frac{1}{2}(A - A')$:

$$A - A' = \begin{bmatrix} 3-3 & 2-4 & 5-0 \\ 4-2 & 1-1 & 3-6 \\ 0-5 & 6-3 & 7-7 \end{bmatrix} = \begin{bmatrix} 0 & -2 & 5 \\ 2 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix}$$

$$Q = \frac{1}{2}\begin{bmatrix} 0 & -2 & 5 \\ 2 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & \frac{5}{2} \\ 1 & 0 & -\frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 0 \end{bmatrix}$$

4. Writing $A = P + Q$:

$$\begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix} = \begin{bmatrix} 3 & 3 & \frac{5}{2} \\ 3 & 1 & \frac{9}{2} \\ \frac{5}{2} & \frac{9}{2} & 7 \end{bmatrix} + \begin{bmatrix} 0 & -1 & \frac{5}{2} \\ 1 & 0 & -\frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 0 \end{bmatrix}$$

Part (ii): Let $A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}$.

1. Find the transpose A':

$$A' = \begin{bmatrix} 2 & 7 & 1 \\ 4 & 3 & -2 \\ -6 & 5 & 4 \end{bmatrix}$$

2. Compute the symmetric matrix $P = \frac{1}{2}(A + A')$:

$$A + A' = \begin{bmatrix} 2+2 & 4+7 & -6+1 \\ 7+4 & 3+3 & 5-2 \\ 1-6 & -2+5 & 4+4 \end{bmatrix} = \begin{bmatrix} 4 & 11 & -5 \\ 11 & 6 & 3 \\ -5 & 3 & 8 \end{bmatrix}$$

$$P = \frac{1}{2}\begin{bmatrix} 4 & 11 & -5 \\ 11 & 6 & 3 \\ -5 & 3 & 8 \end{bmatrix} = \begin{bmatrix} 2 & \frac{11}{2} & -\frac{5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 4 \end{bmatrix}$$

3. Compute the skew-symmetric matrix $Q = \frac{1}{2}(A - A')$:

$$A - A' = \begin{bmatrix} 2-2 & 4-7 & -6-1 \\ 7-4 & 3-3 & 5-(-2) \\ 1-(-6) & -2-5 & 4-4 \end{bmatrix} = \begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix}$$

$$Q = \frac{1}{2}\begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{3}{2} & -\frac{7}{2} \\ \frac{3}{2} & 0 & \frac{7}{2} \\ \frac{7}{2} & -\frac{7}{2} & 0 \end{bmatrix}$$

4. Writing $A = P + Q$:

$$\begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & \frac{11}{2} & -\frac{5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 4 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{3}{2} & -\frac{7}{2} \\ \frac{3}{2} & 0 & \frac{7}{2} \\ \frac{7}{2} & -\frac{7}{2} & 0 \end{bmatrix}$$

Final Answer

(i) $\begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix} = \begin{bmatrix} 3 & 3 & \frac{5}{2} \\ 3 & 1 & \frac{9}{2} \\ \frac{5}{2} & \frac{9}{2} & 7 \end{bmatrix} + \begin{bmatrix} 0 & -1 & \frac{5}{2} \\ 1 & 0 & -\frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 0 \end{bmatrix}$

(ii) $\begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & \frac{11}{2} & -\frac{5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 4 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{3}{2} & -\frac{7}{2} \\ \frac{3}{2} & 0 & \frac{7}{2} \\ \frac{7}{2} & -\frac{7}{2} & 0 \end{bmatrix}$

Question 31

For what value of x, is $A = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ x & -3 & 0 \end{bmatrix}$ a skew symmetric matrix?

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: $A = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ x & -3 & 0 \end{bmatrix}$ is a skew-symmetric matrix

To Find: Value of x

Concept / Formula Used: $A \quad A' = -A \implies a_{ij} = -a_{ji} \quad i, j$

Solution:

By definition, for any skew-symmetric matrix, off-diagonal elements satisfy $a_{ij} = -a_{ji}$.

Comparing element a_{31} with element a_{13}:

$$\begin{aligned} a_{31} &= -a_{13} \\ x &= -(-2) \\ x &= 2 \end{aligned}$$

Verifying with element a_{32} and element a_{23}:

$$\begin{aligned} a_{32} &= -a_{23} \\ -3 &= -(3) \quad \text{(which is consistent)} \end{aligned}$$

Final Answer

$x = 2$

Question 32

If the matrix $A = \begin{bmatrix} a & 6 & b \\ 2c & x+1 & 8 \\ -1 & -8 & y-3 \end{bmatrix}$ is a skew symmetric matrix, then find the value of $ab + bc - xy$.

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: $A = \begin{bmatrix} a & 6 & b \\ 2c & x+1 & 8 \\ -1 & -8 & y-3 \end{bmatrix}$ is a skew-symmetric matrix

To Find: Value of $ab + bc - xy$

Concept / Formula Used: $a_{ii} = 0 \quad a_{ij} = -a_{ji} \quad i \neq j$

Solution:

1. Setting diagonal elements to zero ($a_{ii} = 0$):

$$\begin{aligned} a_{11} = 0 &\implies a = 0 \\ a_{22} = 0 &\implies x + 1 = 0 \implies x = -1 \\ a_{33} = 0 &\implies y - 3 = 0 \implies y = 3 \end{aligned}$$

2. Using $a_{ij} = -a_{ji}$ for off-diagonal elements:

$$\begin{aligned} a_{13} &= -a_{31} \implies b = -(-1) \implies b = 1 \\ a_{21} &= -a_{12} \implies 2c = -(6) \implies 2c = -6 \implies c = -3 \\ a_{23} &= -a_{32} \implies 8 = -(-8) \implies 8 = 8 \quad \text{(Consistent)} \end{aligned}$$

3. Evaluating each term in the required expression $ab + bc - xy$:

$$\begin{aligned} ab &= (0)(1) = 0 \\ bc &= (1)(-3) = -3 \\ xy &= (-1)(3) = -3 \end{aligned}$$

4. Combining the terms:

$$\begin{aligned} ab + bc - xy &= 0 + (-3) - (-3) \\ &= -3 + 3 \\ &= 0 \end{aligned}$$

Final Answer

$ab + bc - xy = 0$